

Variations in Trading Activity, Costly Arbitrage, and Cryptocurrency Returns

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Abstract

Motivated by the cross-sectional and temporal fluctuations in their trading activity, digital assets provide a fruitful environment to investigate the impact of trading activity on expected returns. Portfolio-level analyses and cryptocurrency-level cross-sectional regressions demonstrate a negative and significant relation between expected returns and variability of dollar trading volume and turnover. This effect is robust to controls for beta, size, momentum, network premia, lottery demand, liquidity, idiosyncratic, volatility, and downside risks. The effect of trading activity variation is more pronounced among cryptocurrencies that are costlier to arbitrage. A zero-cost strategy exploiting this predictability survives transaction costs, various size and liquidity screens.

Keywords: Cryptocurrency, trading activity, liquidity, volume, costly arbitrage

JEL: G10, G11, G12

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1 Introduction

It is generally accepted that liquidity is important for understanding asset returns. The existing evidence on liquidity primarily focuses on arguably the most studied and most liquid markets — the equity market in the United States and the global foreign exchange market. With the rising prices and public awareness of Bitcoin, cryptocurrency investments have been attracting increasing attention and trading volumes. Nevertheless, liquidity in the cryptocurrency space is still not well understood due to limited transparency, decentralized structure, and lax regulatory oversight of the crypto landscape. The recent collapse of crypto giant FTX and the evidence of cryptocurrency crashes highlight the importance of understanding time-variation in liquidity in the cryptocurrency market. The main contribution of this paper is to examine the effect of liquidity fluctuations on expected returns. It is reasonable to conjecture that the variations in liquidity should yield particularly powerful effects in the cryptocurrency market, which is commonly regarded as less liquid.

Our empirical evidence focuses on a moderately large cross-section of cryptocurrencies traded against the U.S. Dollar (USD) from September 1, 2017, to September 1, 2022. We aggregate the daily price and volume data across multiple centralized exchanges. In our empirical analysis, we employ two measures of trading activity — dollar trading volume and turnover — to proxy for the liquidity of individual cryptocurrencies. Given the extreme cross-sectional and temporal heterogeneity in trading activity across digital assets, we employ the coefficient of variation to quantify the variability of volume and turnover. We next implement portfolio-level analyses and cryptocurrency-level cross-sectional regressions to study the relationship between expected returns and the coefficient of variation measures.

Univariate portfolio analyses show that cryptocurrencies with high (low) trading activity variation earn low (high) future raw and risk-adjusted returns. The excess return difference between quintile portfolios with the lowest and highest coefficient of variation of volume (turnover) is economically large and statistically significant regardless of value- or equal-weighted constructions, though the magnitudes are larger in the case of equal-weighting. The zero-cost strategy generates a positive and significant alpha after adjusting for the market, size, and momentum factors of [Liu et al. \(2022\)](#). The results are also robust to using various windows ranging from 7 to 30 days to calculate the coefficient of varia-

tion measures. To better understand the nature of this relationship, we look at the average statistics of cryptocurrencies in quintile portfolios sorted on volume (turnover) variability. The cryptocurrencies with the higher coefficient of variation tend to be smaller, less liquid, and more volatile as well as have higher idiosyncratic and downside risks. Thus, the negative and strong relationship between variations in trading activity and expected returns surprisingly contradicts the risk-return trade-off.

Furthermore, we demonstrate that the effect of volume (turnover) variability persists after controlling for a number of cryptocurrency characteristics in the bivariate portfolio analyses and the cross-sectional regressions of [Fama and MacBeth \(1973\)](#). We first demonstrate that the risk associated with fluctuations in liquidity is not explained by established cryptocurrency anomalies — market beta, size, and momentum ([Liu et al., 2022](#)) and the network premia captured by the network-to-market ratio ([Cong et al., 2021](#); [Liebi, 2022](#)). Furthermore, the negative relationship between the coefficient of variation of trading activity and expected returns is robust to the level of volume and turnover, volume shocks ([Babiak et al., 2022](#)), and liquidity measures — the bid-ask spread ([Corwin and Schultz, 2012](#); [Abdi and Ranaldo, 2017](#)) and (realized) Amihud illiquidity measures ([Amihud, 2002](#); [Ranaldo and de Magistris, 2022](#)). Quantitatively, we find that liquidity proxies partially capture the observed effect, though it remains economically and statistically significant.

Next, motivated by the idea that investors may have a preference for cryptocurrencies with lottery-like payoffs, we investigate the significance of lottery demand in explaining the effect of liquidity variability. For this reason, we compute a common proxy for lottery preferences — a maximum daily return (MAX) ([Bali et al., 2011](#)). Controlling for the MAX effect diminishes the average return differences between cryptocurrencies with the lowest and highest liquidity variability in double sorts and reduces the negative estimate in the cross-sectional regressions. Nevertheless, both the average return and the slope coefficient remain statistically significant at the conventional confidence levels. By construction, high MAX cryptocurrencies will also have high idiosyncratic and realized volatility. We calculate both volatility measures following [Ang et al. \(2006, 2009\)](#). Given the recent decline in prices of Bitcoin and other cryptocurrencies, we also extend our analysis to account for the downside risk effect in [Zhang et al. \(2021\)](#) and calculate value-at-risk and expected short-fall characteristics. We demonstrate that accounting for volatility and downside risks leads

to the same conclusion: the effect of trading activity variability survives in the portfolio analyses and the cross-sectional regressions.

We next provide a plausible explanation of the anomalous effect of variations in trading activity. Several observations point towards the limits-to-arbitrage theory behind our results. Exploiting the liquidity variability effect requires short-selling cryptocurrencies with the highest coefficient of variation of volume (turnover). These digital assets appear to be smaller, less liquid, more volatile, and prone to downside risks. Thus, it may be costly to engage in their short-selling. Furthermore, these cryptocurrencies have a higher idiosyncratic risk, which is regarded as the primary arbitrage cost by the extant literature. To test this hypothesis, we create an arbitrage index capturing different dimensions of arbitrage costs via idiosyncratic volatility and three liquidity proxies. The independent double sorts based on the arbitrage index and the coefficient of variation confirm that the mispricing is concentrated in cryptocurrencies with the highest arbitrage costs.

Our empirical results are based on the screen, which chooses the top 100 cryptocurrencies based on market capitalization each day. This ensures that the results are not driven by tiny tokens. In robustness checks, we confirm that our results are robust to alternative screens including crypto assets with market values above \$1 (\$2) million following [Cong et al. \(2021\)](#) and [Liu et al. \(2022\)](#) and excluding the least liquid tokens similarly to [Zhang et al. \(2021\)](#). In yet another exercise, we check that profitability of the zero-cost strategy exploiting the volume (turnover) variability effect is robust to simplistic transaction costs.

Our paper contributes to the growing research on the cross-sectional predictors of cryptocurrency returns. [Liu et al. \(2020\)](#), [Shen et al. \(2020\)](#), and [Liu et al. \(2022\)](#) propose the market, size, and momentum factors. [Cong et al. \(2021\)](#) and [Liebi \(2022\)](#) augment the three-factor model to include the value premium. A number of additional studies highlight the network effects to explain cryptocurrency returns ([Li et al., 2020](#); [Bhambhwani et al., 2021](#); [Liu and Tsyvinski, 2021](#)). The literature also proposes various factors affecting cryptocurrency returns including investor attention ([Wang and Vergne, 2017](#); [Corbet et al., 2022](#)), downside risk ([Dobrynskaya, 2020](#); [Zhang et al., 2021](#)), idiosyncratic volatility ([Zhang and Li, 2020](#)), lottery demand ([Li et al., 2021](#)), and higher moments ([Jia et al., 2021](#)), among others. In addition to observable risk factors, [Babiak and Bianchi \(2021\)](#) and [Borri](#)

[et al. \(2022\)](#) propose a latent factor approach to explain the cross-section of cryptocurrency returns. We contribute to this literature by investigating the role of the second moment of liquidity, proxied by the two measures of trading activity — trading volume and turnover.

Empirically, our paper is motivated by the studies that analyze fluctuations in liquidity in different markets. On the methodological side, our paper is closest to [Chordia et al. \(2001\)](#) studying variations in trading activity in the equity market. We contribute to this literature by providing a comprehensive investigation of the effect of trading activity fluctuations on the cross-section of cryptocurrency returns. In our focus on trading volume, our paper is also related to [Babiak et al. \(2022\)](#). The authors show that returns from liquidity provision, proxied by the returns of a short-term reversal strategy ([Nagel, 2012](#)), are concentrated in cryptocurrencies with low trading volume. Our paper is different from their analysis as we investigate the information content of the second moment of the trading volume. In relation to liquidity in cryptocurrency markets, [Brauneis et al. \(2021\)](#) provides a horse race of various liquidity proxies in their ability to predict the prices of the two cryptocurrencies BTC and ETH. We differ from this study by examining the cross-sectional predictability of the second moment of liquidity. To our knowledge, [Leirvik \(2022\)](#) is the only study focusing on the effect of liquidity volatility. Our paper is distinct from this study because we focus on asset pricing implications for a large cross-section instead of the five biggest cryptocurrencies. Furthermore, our key focus is trading volume and turnover instead of the bid-ask spread employed by [Leirvik \(2022\)](#).

Our paper is connected to the literature on the limits of arbitrage and price formation in the cryptocurrency markets. [Makarov and Schoar \(2020\)](#) demonstrate persistent deviations in Bitcoin prices across exchanges and suggest that there are significant barriers to exploiting these arbitrage opportunities.¹ [Wang et al. \(2022\)](#) investigate cyclic arbitrages in decentralized exchanges. Similarly to this literature, our results suggest limits-to-arbitrage as the likely explanation of mispricing documented in our paper. The recent literature also aims to understand the economics of cryptocurrency trading through the lens of theoretical models including [Sockin and Xiong \(2018\)](#), [Cong et al. \(2021\)](#), and [Cong et al. \(2021\)](#). Our contribution to this strand is to empirically demonstrate how fluctuations in trading

¹In contrast to the costly arbitrage interpretation, [Borri and Shakhnov \(2022\)](#) provide a risk-based explanation of Bitcoin price differences. They show that Bitcoin prices for more expensive pairs depreciate more in bad times, when aggregate liquidity is low, and hence are riskier.

activity predict future cryptocurrency prices.

2 Trading activity and expected cryptocurrency returns

2.1 Data

We retrieve daily data on prices and trading volume from *CryptoCompare* and on-chain activity from *IntoTheBlock*. The full sample covers the period from June 26, 2014, to September 1, 2022, with a day defined at a start time of 00:00:00 UTC. The prices and volume are aggregated across more than 80 different centralized exchanges.² The aggregation across different exchanges uses a volume-weighted scheme, which accounts for the exchange-specific trading activity. Thus, larger and more established exchanges tend to have relatively more weight in the aggregation of prices and trading volume compared to smaller and peripheral exchanges. This volume-weighted scheme is standard in the literature (*Liu et al., 2022*). The cryptocurrencies use USD as the quote currency and are included in the sample if all data from *CryptoCompare* and *IntoTheBlock* are available.

We introduce various filters to mitigate the impact of erratic or suspicious trading activity. First, we exclude observations with a zero trading volume or a zero price. Second, for each day we compute the ratio of trading volume to the market capitalization of each cryptocurrency and exclude observations with a ratio greater than 1. This simple filter screens out cryptocurrencies with potentially fake volume. Third, we exclude (1) all cryptocurrencies that are backed by or track the price of gold or any precious metal; (2) wrapped coins (e.g., WBTC), because they represent copies of existing tokens; (3) stablecoins, including those that are centralized (e.g., USDT, USDC) and algorithmically stabilized (e.g., DAI, UST) for any fiat currencies; and (4) synthetic derivatives (e.g., stETH, stSOL), as they track the value of an existing cryptocurrency. Our screening follows the classification provided by *CoinMarketCap*. Appendix A presents a summary of the procedure and additional details of the filters implemented by *CryptoCompare* to mitigate the impact of fraudulent trading activity and malfunctioning API for individual exchanges.

These steps mitigate the effect of fake volume and substantially reduce the exposure of the empirical analysis to misreported trading activity for some exchanges. The two

²We include the exchanges with the ranking from AA to B as provided by *CryptoCompare*. One can view the exchange grades on the company website at <https://www.cryptocompare.com/exchanges/overview>.

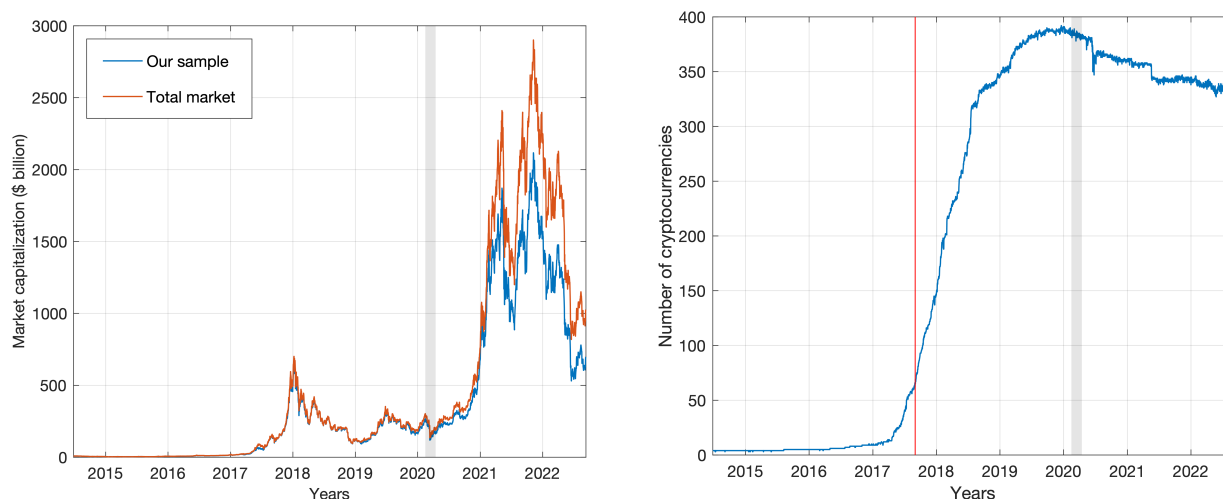


Figure 1. Market capitalization and the cross-section of cryptocurrencies.

This figure compares the market value of cryptocurrencies in our sample and the total market capitalization (a left panel) and illustrates the number of cryptocurrencies in our sample over time (a right panel). The grey area denotes the COVID-19 recession. The vertical red line indicates the start of the studied period.

comments are noteworthy. First, fake trading is typically performed via crypto-to-crypto trading on individual and relatively smaller exchanges that inflate trading volume in order to attract Initial Coin Offering's (ICO) listings or to manipulate the market (Li et al., 2018; Cong et al., 2021). We consider the cryptocurrencies traded against a fiat currency and then aggregate the price and volume data over a large cross-section of exchanges. Therefore, the risk that the manipulation of relatively small exchanges could affect the overall market activity is substantially mitigated. Second, the fact that we focus on transactions taking place on regular trading exchanges should mitigate the concern that market activity is primarily driven by illegal activities. The latter typically do not take place on registered centralized or decentralized exchanges but through peer-to-peer transactions on the blockchain (Foley et al., 2019; Griffin and Shams, 2020).

This screening process yields an unbalanced panel of 405 cryptocurrencies. Figure 1 illustrates the market value of cryptocurrencies in our sample relative to the total market capitalization. Although our sample covers a smaller number of cryptocurrencies than currently available – there are more than 19,000 different tokens at the time of writing according to CoinMarketCap – our coverage of the market value is substantial. In relative terms, our dataset constitutes 95% of the total market value from 2014 until late 2019 and above 70% at the end of the period under consideration. As shown in Figure 1, the initial size of the cross-section is small with a handful of available cryptocurrencies before 2017.

This number rapidly increases from around 50 pairs in August 2017 to around 400 pairs by the end of 2019 and then slightly decreases to around 300 by the end of the sample.

2.2 Final sample and variable definitions

In our empirical analysis, we focus on the sample of 405 cryptocurrencies covering the period from September 1, 2017, to September 1, 2022. We choose the starting date of September 1, 2017, because the number of available cryptocurrencies exceeds 50 at that time, which enables us to perform univariate and bivariate portfolio sorts.

The key focus of our empirical investigation is the variation of trading activity and its relation with future returns. We focus on two proxies of trading activity: dollar trading volume defined as the total dollar value of coins transferred across wallets within and across centralized exchanges, and turnover defined as the ratio of dollar trading volume to the current supply ([Datar et al., 1998](#)). The current supply is the total number of coins or tokens that have been mined and includes all cryptocurrencies in public hands, either circulating in the market or locked/vested. Following [Cong et al. \(2021\)](#), we compute the average daily volume (VOL) and turnover (TURN) over the past seven days for each cryptocurrency. We calculate the coefficient of variation of volume (turnover), which is defined as the ratio between the standard deviation and the mean of daily dollar trading volume (turnover) over the past k days and denoted by $CVVOL_k$ ($CVTURN_k$). We consider four choices of $k = 7, 14, 21, 30$ days and verify the robustness of our results to different lengths of a rolling window.

A significant relation between variations in trading activity and expected cryptocurrency returns can be explained by other fundamental risks or cryptocurrency-specific attributes that highly correlate with changes in trading volume and turnover. We account for this possibility by controlling for additional characteristics of digital assets that have been shown to predict future returns by the existing literature. Table 1 provides the description of cryptocurrency characteristics.

[Liu et al. \(2022\)](#) propose that three risk factors – market, size, and momentum – explain the cross-sectional expected cryptocurrency returns. Motivated by this evidence, we estimate the CAPM regression of daily cryptocurrency returns on the value-weighted market returns using a 30-day rolling window ([Lewellen and Nagel, 2006](#)). For each cryptocur-

Table 1. Cryptocurrency characteristic definitions.

This table describes the cryptocurrency variables used in the empirical analysis. The first (second) column shows the name (definition) of each characteristic.

Characteristic	Definition
VOL	Average daily dollar trading volume over the past seven days.
TURN	Average daily turnover over the past seven days.
CVVOL k	The ratio between standard deviation and mean of volume over the past k days.
CVTURN k	The ratio between standard deviation and mean of turnover over the past k days.
Alpha & Beta	The intercept and the slope coefficient in the regression of cryptocurrency returns on the market return using daily data over the past 30 days.
SIZE	Last-day market capitalization, measured as the product of the current supply and the current USD price.
MOM k	Past k -day momentum, measured as the cumulative return over the past k days.
BM	The ratio between the cumulative number of unique addresses and the current supply, multiplied by the current USD price of a particular digital asset.
VSHOCK	Log log deviation of trading volume from its trend computed over the past 30 days. The log standard deviation calculated over the same rolling window is used to standardize the estimates due to cross-sectional imbalances.
BA	The average of two synthetic bid-ask spreads of Corwin and Schultz (2012) and Abdi and Ranaldo (2017) based on daily open, high, low, and close prices.
ILLIQ	The ratio between the sum of absolute daily returns and the average daily volume over the past seven days.
RILLIQ	The ratio between the realized volatility of daily returns and the average daily volume over the past seven days.
MAX k	A maximum daily return over the past k days.
IVOL	The volatility of residuals from the regression of daily returns on the market return using daily data over the past 30 days.
RVOL k	The volatility of daily returns over the past k days.
NEW	The number of unique addresses that appeared for the first time in a transaction of the native coin in the network.
ACT	The number of unique addresses that were active in the network either as a sender or receiver. Only addresses that were active in successful transactions are counted.
VaR95 & ES95	The fifth percentile and the mean daily return below the fifth percentile of daily returns over the past 90 days.

rency, we save conditional alpha (ALPHA) and beta (BETA) from these short-window regressions. We compute each cryptocurrency's market capitalization (SIZE) as the product of the current supply and the current USD price. We also calculate past k -day momentum (MOM k) as the cumulative return over the past $k = 7, 14, 21, 30$ days. Recently, [Cong et al. \(2021\)](#) and [Liebi \(2022\)](#) show that blockchain fundamentals – the fundamental-to-market

ratio called the crypto value factor – determine crypto asset prices. Thus, we compute the network-to-market value (BM) of the individual cryptocurrency as the ratio of a cumulative number of unique addresses over the current supply, which is multiplied by the current USD price.

Next, we control for a set of liquidity measures. Using daily open, high, low, and close prices, we construct two approximations of the bid-ask spread following the methodology of [Corwin and Schultz \(2012\)](#) and [Abdi and Ranaldo \(2017\)](#). We then take a simple average of the two proxies to construct the bid-ask spread (BA). Following [Amihud \(2002\)](#), we construct the Amihud illiquidity measure, defined as the ratio between the sum of absolute daily returns and the average daily volume over the past seven days (ILLIQ). Next, following [Ranaldo and de Magistris \(2022\)](#) who develop a refinement of the standard Amihud illiquidity measure, we compute the realized Amihud illiquidity (RILLIQ) as the ratio between realized volatility and the average daily volume over the past seven days. [Babiarz et al. \(2022\)](#) demonstrate that returns from liquidity provision are mainly concentrated in cryptocurrencies with lower levels of trading activity. Motivated by this evidence and the information content of abnormally low volume in traditional currency markets ([Cespa et al., 2022](#)), we control for volume changes by calculating volume shocks (VSHOCK), defined as the (log) deviation of volume from its trend – the log average daily volume over the past 30 days.

Furthermore, we uncover a potential connection between the variations in trading activity and lottery demand, idiosyncratic and realized volatility, and downside risk. [Bali et al. \(2011\)](#) demonstrate the importance of lottery-like payoffs in asset pricing. Adopting their proxies of lottery demand, we compute the maximum daily return over the last $k = 7, 14, 21, 30$ days (MAX k). Following [Ang et al. \(2006\)](#), we calculate idiosyncratic volatility (IVOL) as the standard deviation of residuals from the regression used to obtain conditional alphas and betas. We construct realized volatility (RVOL k) as the standard deviation of daily realized returns during the past k days. [Zhang et al. \(2021\)](#) document a positive relation between downside risk and future returns. Thus, we calculate value-at-risk and expected shortfall at 95% (VaR95 and ES95) estimated based on daily returns over a 90-day rolling window.

Finally, a large strand of the academic literature suggests that cryptocurrencies are primarily driven by psychological factors. As prices of digital assets move up and down, trading activity may reflect shifts in the demand of retail and institutional investors. The clientele hypothesis posits that assets with greater investor following should earn higher expected returns ([Merton, 1987](#)). Although we do not have the data on cryptocurrency ownership, we partially explore this hypothesis by exploiting the blockchain data on new and active addresses similar to [Chordia et al. \(2001\)](#). For each day, we compute the number of unique addresses that appeared for the first time in a transaction of the native coin in the network (NEW). We further calculate unique addresses that were active in the network either as a sender or receiver (ACT).

2.3 Portfolio formation

In our empirical investigation, we implement a variety of single-sorted and double-sorted portfolios. We consider value-weighted (based on market capitalization) and equal-weighted portfolios formed daily using various cryptocurrency characteristics. In our analysis, we consider only those cryptocurrencies that on a particular day have non-zero market capitalization and non-missing characteristic(s) used for portfolio selection as well as a non-missing return on the next trading data. In doing so, we ensure that we are able to form value- and equal-weighted portfolios at the end of the current day and track the returns of each portfolio on the following day. Furthermore, each day we select the top 100 cryptocurrencies based on the current market capitalization. This implies that assets employed in portfolio construction are dynamically adjusted to exclude the smallest coins. In robustness checks, we implement alternative screens that include cryptocurrencies with a market capitalization larger than \$1 million similar to [Liu et al. \(2022\)](#) and [Cong et al. \(2021\)](#), or exclude the least liquid cryptocurrencies according to the Amihud illiquidity measure similar to [Zhang et al. \(2021\)](#). As shown in Section 3.7, our key conclusions remain unaltered when we apply different portfolio screens.

2.4 Univariate portfolio analysis

To begin our analysis, we form quintile portfolios based on the coefficient of variation of volume (turnover). We implement single-sorted portfolios on a daily basis between September 1, 2017, and September 1, 2022. We further look at the long-short portfolio

buying and selling quintiles of cryptocurrencies with the lowest and highest variations of trading volume (turnover). Table 2 presents summary statistics of the value- and equal-weighted returns of CCVOL portfolios. Panels A, B, C, and D showcase portfolios based on the coefficient of variation calculated over 7-, 14-, 21-, and 30-day rolling windows.

As shown in Panel A, a value-weighted portfolio combining cryptocurrencies with the lowest CVVOL values earns a monthly average return of -1.68%. The mean returns tend to decrease when we move from the lowest to the highest CVVOL portfolios. In particular, the fourth and fifth quintiles generate -8.61% and -18.04% per month. The value-weighted average raw return difference between the low CVVOL portfolio and the high CVVOL portfolio is 16.36% with a significantly positive Newey and West (1986) adjusted t-statistic of 5.08. This illustrates that cryptocurrencies with higher variation in trading activity earn significantly lower expected returns. The difference in average returns translates into the (almost) monotonically decreasing pattern in Sharpe ratios. For example, the lowest CVVOL portfolio produces a Sharpe ratio of -0.06 per month, whereas the two portfolios with the highest CVVOL measures have risk-adjusted performances of -0.27 and -0.44.

Next, we examine whether standard asset pricing models can explain the portfolio returns sorted on the variations in trading volume. Table 2 presents the monthly alphas after adjusting for the market, size, and momentum factors of Liu et al. (2022). Panel A for value-weighted portfolios indicates that the lowest CVVOL portfolio generates an alpha of -7.61%, whereas the highest CVVOL portfolio produces an alpha of -27.80% per month. Moreover, the three-factor alphas exhibit the same pattern as in returns, with the highest (lowest) alphas for the lowest (highest) CVVOL portfolio and a sharp decline in values starting from the third quintile. Most importantly, the alpha difference of quintiles with extreme variations in trading activity equals 20.19% on a monthly basis, which is both economically and statistically significant. The right part of Panel A presents results for the equal-weighted portfolios and our findings are qualitatively similar. Quantitatively, one can observe a stronger underperformance of quintiles comprising cryptocurrencies with a higher coefficient of variation measures. The larger spread in raw and risk-adjusted excess returns is visible as well.

Panels B, C, and D in Table 2 show that our findings are robust to applying alternative

Table 2. Returns on CVVOL portfolios.

This table reports summary statistics of value- and equal-weighted quintile portfolios formed daily based on the coefficient of variation of volume between September 1, 2017, and September 1, 2022. The table also reports the returns on the long-short portfolio buying and selling quintiles of cryptocurrencies with the lowest and highest variations in trading volume. Panels A, B, C, and D present the returns on portfolios based on the coefficient of variation measures calculated over the 7-, 14-, 21-, and 30-day rolling windows, respectively. Summary statistics include the mean, standard deviation, Sharpe ratio, alpha (reported on a monthly basis), and skewness (based on daily returns). Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the long-short portfolio.

	Value-Weighted Portfolios						Equal-Weighted Portfolios					
	Mean	Std	SR	Skew	alpha	t-stat	Mean	Std	SR	Skew	alpha	t-stat
Panel A: CVVOL7												
Low	-1.68	28.37	-0.06	-1.36	-7.61	-4.61	-0.37	28.51	-0.01	-1.36	-9.07	-4.93
2	0.82	27.50	0.03	-1.11	-4.41	-2.55	-3.87	29.05	-0.13	-1.44	-12.46	-6.79
3	-1.90	29.57	-0.06	-0.94	-8.37	-4.28	-3.77	29.13	-0.13	-1.22	-12.82	-6.63
4	-8.61	32.31	-0.27	-0.61	-14.31	-5.24	-9.39	28.97	-0.32	-1.43	-17.57	-7.91
High	-18.04	40.92	-0.44	-0.07	-27.80	-8.41	-19.08	29.52	-0.65	-0.83	-28.20	-12.87
LMH	16.36	35.93	0.46	-0.32	20.19	6.25	18.71	17.89	1.05	0.13	19.13	10.08
t-stat	5.08						9.57					
Panel B: CVVOL14												
Low	0.08	27.04	0.00	-0.93	-5.66	-4.23	-2.48	27.97	-0.09	-1.35	-10.74	-6.31
2	2.08	28.25	0.07	-0.67	-2.62	-1.49	-2.94	29.05	-0.10	-1.41	-11.72	-6.32
3	-5.20	29.53	-0.18	-0.63	-10.32	-4.72	-3.58	29.45	-0.12	-1.21	-12.15	-5.99
4	-6.06	35.07	-0.17	-1.46	-11.54	-4.02	-8.20	29.08	-0.28	-1.43	-16.39	-7.35
High	-17.33	38.50	-0.45	0.02	-26.31	-7.41	-19.41	29.49	-0.66	-0.92	-28.97	-12.69
LMH	17.41	32.89	0.53	-1.04	20.65	5.47	16.93	17.06	0.99	0.08	18.23	9.34
t-stat	4.66						8.72					
Panel C: CVVOL21												
Low	-0.10	26.44	0.00	-1.07	-5.27	-3.76	-2.12	27.75	-0.08	-1.35	-10.52	-6.16
2	1.22	27.53	0.04	-0.91	-3.02	-1.79	-3.05	28.74	-0.11	-1.33	-11.46	-6.51
3	-6.08	30.55	-0.20	-0.54	-11.77	-4.78	-4.65	29.64	-0.16	-1.19	-13.43	-6.42
4	-6.54	35.18	-0.19	-0.87	-13.54	-5.00	-7.01	29.06	-0.24	-1.39	-14.96	-6.65
High	-16.85	38.06	-0.44	-0.01	-26.53	-7.55	-19.79	29.23	-0.68	-1.07	-29.37	-13.43
LMH	16.75	31.31	0.54	-1.16	21.25	5.58	17.66	16.48	1.07	0.21	18.85	10.10
t-stat	4.43						9.67					
Panel D: CVVOL30												
Low	1.24	25.66	0.05	-0.73	-3.64	-3.20	-1.46	27.39	-0.05	-1.32	-9.68	-5.82
2	-2.69	28.51	-0.09	-1.04	-7.47	-4.33	-3.78	29.23	-0.13	-1.23	-12.44	-6.79
3	-4.88	30.58	-0.16	-0.92	-10.70	-4.76	-5.84	29.84	-0.20	-1.45	-14.45	-6.96
4	-7.47	35.95	-0.21	-0.71	-15.42	-5.47	-7.79	28.85	-0.27	-1.21	-15.99	-7.91
High	-16.02	38.22	-0.42	-0.11	-24.86	-7.03	-18.09	29.16	-0.62	-1.10	-27.38	-11.55
LMH	17.26	33.19	0.52	-0.40	21.21	5.74	16.63	16.59	1.00	-0.11	17.70	9.05
t-stat	4.76						8.69					

choices of rolling window lengths. Indeed, we observe a monotonically declining pattern in the raw and risk-adjusted average returns and a significant gap in the performance of portfolios with extreme variations in trading volume. The univariate portfolio analysis reveals two main conclusions. First, the finding that the cryptocurrencies with higher variability of trading activity earn lower expected returns cannot be explained by the market,

Table 3. Returns on CVTURN portfolios.

This table reports summary statistics of value- and equal-weighted quintile portfolios formed daily based on the coefficient of variation of turnover between September 1, 2017, and September 1, 2022. The table also reports the returns on the long-short portfolio buying and selling quintiles of cryptocurrencies with the lowest and highest variations in turnover. Panels A, B, C, and D present the returns on portfolios based on the coefficient of variation measures calculated over the 7-, 14-, 21-, and 30-day rolling windows, respectively. Summary statistics include the mean, standard deviation, Sharpe ratio, alpha (reported on a monthly basis), and skewness (based on daily returns). Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the long-short portfolio.

	Value-Weighted Portfolios						Equal-Weighted Portfolios					
	Mean	Std	SR	Skew	alpha	t-stat	Mean	Std	SR	Skew	alpha	t-stat
Panel A: CVTURN7												
Low	-3.62	29.04	-0.12	-1.45	-9.59	-5.43	-2.03	28.81	-0.07	-1.39	-10.80	-5.73
2	1.50	27.42	0.05	-1.11	-3.78	-2.64	-2.76	29.04	-0.10	-1.23	-11.81	-6.66
3	-1.51	28.29	-0.05	-0.95	-6.73	-3.12	-4.82	28.89	-0.17	-1.35	-13.36	-6.30
4	-7.32	32.29	-0.23	-0.89	-12.68	-4.74	-8.67	29.23	-0.30	-1.40	-17.11	-7.73
High	-17.27	37.96	-0.45	-0.20	-26.27	-7.81	-18.20	29.24	-0.62	-0.85	-27.06	-13.14
LMH	13.65	33.23	0.41	-0.08	16.68	4.88	16.18	17.32	0.93	0.27	16.26	8.56
t-stat	3.91						8.47					
Panel B: CVTURN14												
Low	-1.74	28.04	-0.06	-1.08	-7.24	-4.85	-2.90	28.41	-0.10	-1.24	-11.23	-6.44
2	2.38	28.48	0.08	-0.87	-2.26	-1.40	-3.33	29.22	-0.11	-1.49	-12.08	-6.46
3	-5.26	29.52	-0.18	-0.87	-10.94	-4.42	-4.09	28.86	-0.14	-1.27	-12.61	-6.21
4	-3.67	35.55	-0.10	-0.08	-9.84	-3.06	-7.53	29.23	-0.26	-1.48	-15.68	-7.28
High	-13.64	38.37	-0.36	-0.31	-22.64	-6.06	-18.78	29.16	-0.64	-0.89	-28.38	-13.14
LMH	11.90	33.34	0.36	-0.23	15.40	4.04	15.88	16.98	0.94	0.12	17.16	9.31
t-stat	2.88						8.34					
Panel C: CVTURN21												
Low	-1.80	27.61	-0.07	-1.08	-6.66	-5.00	-2.58	28.01	-0.09	-1.25	-10.93	-6.30
2	-0.59	29.33	-0.02	-0.35	-4.54	-2.05	-3.69	29.02	-0.13	-1.43	-12.34	-6.73
3	-1.83	29.24	-0.06	-0.41	-8.64	-3.93	-4.10	29.47	-0.14	-1.26	-12.57	-6.32
4	-5.58	33.63	-0.17	-1.12	-13.21	-4.44	-8.12	28.82	-0.28	-1.37	-16.37	-7.70
High	-14.76	38.33	-0.39	-0.17	-23.26	-6.91	-18.13	29.08	-0.62	-1.05	-27.53	-12.64
LMH	12.96	31.91	0.41	-0.70	16.60	4.69	15.55	16.40	0.95	0.13	16.60	8.94
t-stat	3.43						8.28					
Panel D: CVTURN30												
Low	-2.11	27.29	-0.08	-1.11	-7.11	-5.06	-2.24	28.10	-0.08	-1.39	-10.65	-6.11
2	-1.65	28.79	-0.06	-0.97	-5.79	-3.48	-5.18	28.85	-0.18	-1.11	-13.73	-7.62
3	-4.45	30.29	-0.15	-0.79	-10.51	-4.35	-4.06	29.70	-0.14	-1.55	-12.10	-5.69
4	-5.73	35.20	-0.16	-0.74	-13.49	-4.74	-8.21	28.85	-0.28	-1.23	-16.77	-8.75
High	-14.51	38.08	-0.38	0.27	-24.18	-7.41	-17.27	28.97	-0.60	-1.05	-26.69	-11.33
LMH	12.40	32.66	0.38	-0.68	17.07	5.21	15.03	16.80	0.89	-0.01	16.04	7.81
t-stat	3.43						7.31					

size, or momentum factors proposed by [Liu et al. \(2022\)](#). Second, this evidence is mainly driven by the strong underperformance of cryptocurrencies with extremely high variations in trading volume. Intuitively, the results suggest that investors are willing to pay higher prices for cryptocurrencies with abnormally high increases in trading volume relative to recent trades and, consequently, experience significantly negative raw and risk-adjusted

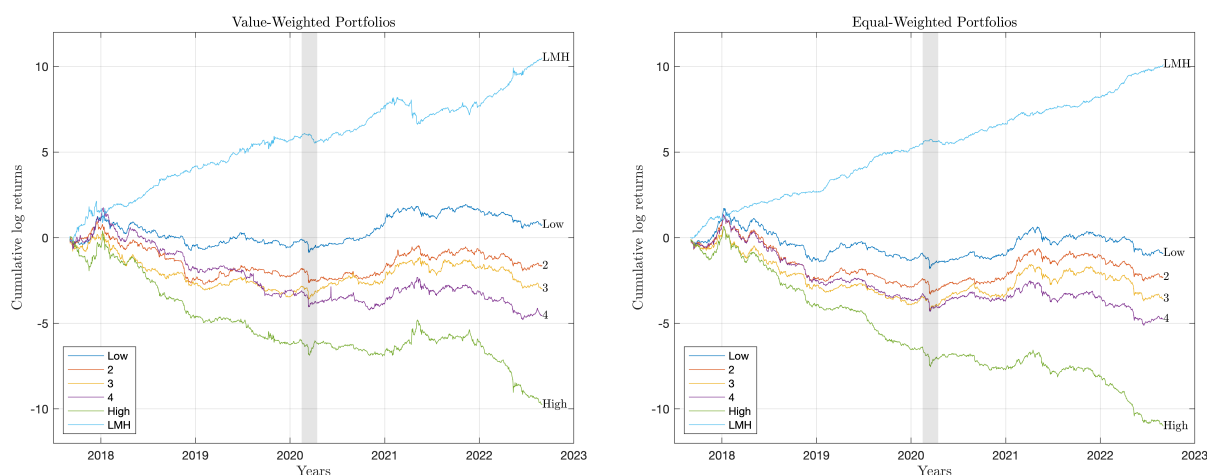


Figure 2. Cumulative returns on CVVOL portfolios.

This figure illustrates cumulative returns on value-weighted (a left panel) and equal-weighted (a right panel) quintile portfolios formed daily based on the coefficient of variation of dollar trading volume between September 1, 2017, and September 1, 2022. This figure also illustrates cumulative returns on the long-short portfolio buying and selling quintiles of cryptocurrencies with the lowest and highest variations of the trading volume. The coefficient of variation measure is calculated over the 30-day rolling window. The grey area denotes the COVID-19 recession.

returns in the future.

Table 3 reports results when using turnover instead of volume as a proxy of trading activity. Panel A reports a monthly value-weighted average return of -3.62% (-17.27%), a monthly Sharpe ratio of -0.12 (-0.45), and a monthly three-factor alpha of -9.59% (-26.27%) for the lowest (highest) CVTURN quintile. This yields an average return difference between the first and fifth quintile portfolios of 13.65% with a t-statistic of 3.91 as well as economically significant risk-adjusted performances. Note that there is a sharp decline in excess returns, Sharpe ratios, and alphas of the fourth and fifth quintiles, which contain cryptocurrencies with the highest CVTURN quantities. The equal-weighted quintile and long-short portfolios exhibit performance metrics of a similar magnitude. Panels B, C, and D report portfolio results for longer rolling window sizes and confirm that our findings remain qualitatively and quantitatively similar. Overall, the presented evidence demonstrates a negative relation between expected cryptocurrency returns and the variability of trading activity as measured by dollar trading volume or turnover.

Figure 2 augments results by illustrating the cumulative returns of CVVOL portfolios for the case of a 30-day rolling window. Consistent with the average returns of quintiles, the cumulative performance exhibits a decreasing pattern moving from cryptocurrencies

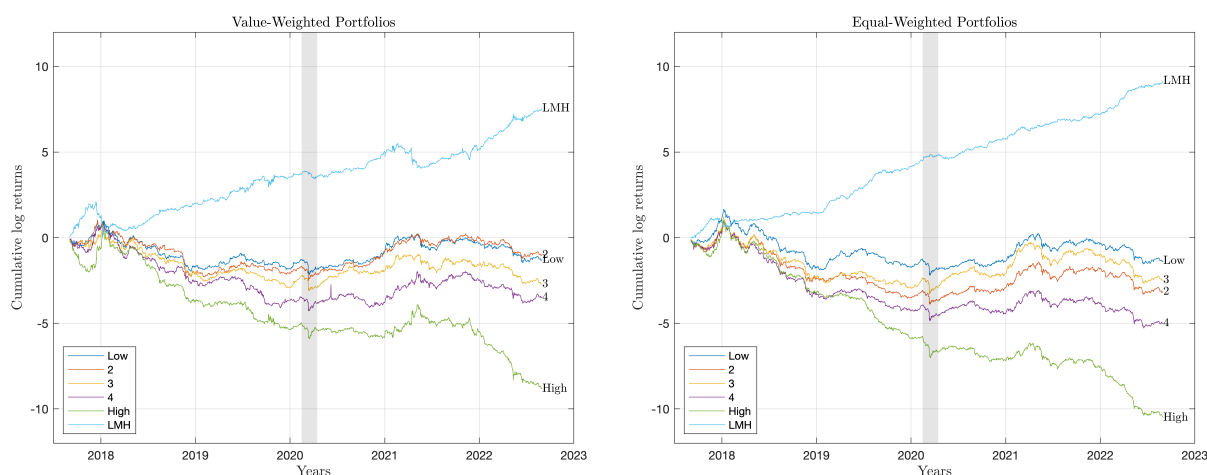


Figure 3. Cumulative returns on CVTURN portfolios.

This figure illustrates cumulative returns on value-weighted (a left panel) and equal-weighted (a right panel) quintile portfolios formed daily based on the coefficient of variation of turnover between September 1, 2017, and September 1, 2022. This figure also illustrates cumulative returns on the long-short portfolio buying and selling quintiles of cryptocurrencies with the lowest and highest variations of turnover. The coefficient of variation measure is calculated over the 30-day rolling window. The grey area denotes the COVID-19 recession.

with the lowest coefficient of variations to those with the highest. The zero-cost strategies exploiting variability in trading volume generate a similar cumulative return of 1000% during the considered period for the value- and equal-weighted schemes. A value-weighted long-short portfolio, however, produces higher volatility and experiences larger declines in bust cycles as indicated by the negative returns during the “crypto-winter” of early 2018, the COVID-19 crash in March 2020, and the Bitcoin meltdown in 2021. Figure 3 demonstrates that the predictive power of turnover variations on future returns is slightly weaker. Indeed, the cumulative performance of an equal-weighted zero-cost strategy diminishes to around 900%, whereas a value-weighted investment produces a cumulative return of around 750% by the end of the sample.

2.5 Average portfolio characteristics

We now examine the properties of cryptocurrencies that belong to different quintiles based on the coefficient of variation measures. This analysis may shed more light on the cryptocurrency-specific characteristics that can potentially explain the significant negative relation between expected returns and variations in trading activity. To save space, we consider the case of CVVOL and CVTURN measures estimated over the 30-day rolling window. To do so, we divide cryptocurrencies into quintiles using the corresponding

metrics, compute the median values of various cryptocurrency-specific characteristics for each quintile, and calculate the time-series averages of the cross-sectional medians.

Table 4 reports the average portfolio characteristics. Focusing on results in Panel A, several observations are noteworthy. First, the average coefficients of variation of dollar trading volume and turnover monotonically increase from the first to the fifth quintile. This pattern is expected because both measures are strongly correlated by construction. The volume shock characteristic is strongly negative (slightly positive) for cryptocurrencies with a low (high) coefficient of variation. This observation indicates that low variability of trading activity is associated with trading volume below its trend, whereas high variability reflects abnormally high trading volume.

Second, one observes a monotonic decline in beta and size as the variation of trading volume increases. For instance, the average market capitalization of cryptocurrencies in the lowest (highest) CCVOL portfolio is around \$364.6 (\$107.7) million. The first quintile has a slight value tilt in beta, however, the average beta is decreasing from 1.04 to 0.95 when moving from the second to the fifth quintile.³ The network-to-market value is also decreasing in CCVOL. Considering that value cryptocurrencies yield higher average returns than growth cryptocurrencies (Liebi, 2022), blockchain fundamentals associated with the value premium may drive a significant portion of our findings. In relation to network characteristics, one also observes that cryptocurrencies with lower (higher) trading variability tend to have a bigger (smaller) number of new and active addresses.

Third, the average raw and risk-adjusted past performance is increasing in the coefficient of variation. That is, the cryptocurrencies with abnormally high trading activity tend to have a strong recent performance. Thus, agents may over-value those assets and are willing to pay more for cryptocurrencies that exhibit extreme variation in trading activity, which leads to lower expected returns in the future. Additionally, the average daily maximum returns increase steadily from the first to the fifth quintile. This measure reflects investors' demand for assets with lottery-like payoffs (Bali et al., 2011).

Fourth, there is a strong increasing pattern in the average liquidity measures across the

³We obtain betas daily using the 30-day rolling window estimation. Due to a short-window regression, these estimates likely provide a noisy estimate of the true betas that could explain a slight drop in the average beta of the lowest coefficient of variation portfolio.

Table 4. Summary statistics for CVVOL and CVTURN portfolios.

This table reports the average characteristics of quintile portfolios formed daily based on the coefficient of variation of volume (Panel A) or turnover (Panel B) between September 1, 2017, and September 1, 2022. The table shows results based on the coefficient of variation measures calculated over the 30-day rolling window. The table presents the time-series averages of the daily median values of various cryptocurrency-specific characteristics for each quintile. The cryptocurrency-specific attributes include the coefficient of variation of volume (CVVOL) and turnover (CVTURN) calculated over the 30-day rolling window; market beta (BETA); market capitalization (SIZE) and trading volume (VOL) expressed in million; past 30-day momentum (MOM), CAPM alpha (ALPHA), book-to-market ratio (BM), and turnover (TURN) expressed in percentage; volume shock (VSHOCK); bid-ask spread (BA); Amihud and realized Amihud illiquidity measures (ILLIQ and RILLIQ); new and active addresses (NEW and ACT); a maximum daily return over the past 30 days (MAX), idiosyncratic volatility (IVOL), realized volatility over the past 30 days (RVOL), value-at-risk (VAR95), and expected shortfall (ES95) expressed in percentage.

Panel A: CVVOL30										
	CVVOL	CVTURN	BETA	SIZE	MOM (%)	ALPHA (%)	BM (%)	VOL	TURN (%)	VSHOCK
Low	0.37	0.35	1.00	364.64	-0.24	-0.25	0.09	1.93	0.67	-0.13
2	0.56	0.53	1.04	260.44	-0.16	-0.17	0.07	0.95	0.32	-0.14
3	0.76	0.71	1.04	161.85	-0.08	-0.09	0.07	0.53	0.24	-0.09
4	1.05	0.97	1.00	119.77	0.05	0.03	0.07	0.34	0.17	0.03
High	1.72	1.62	0.95	107.73	0.17	0.15	0.06	0.20	0.08	0.11
	BA	ILLIQ	RILLIQ	NEW	ACT	MAX (%)	IVOL (%)	RVOL (%)	VAR95 (%)	ES95 (%)
Low	0.04	0.04	0.01	54.40	160.63	12.49	4.05	6.03	10.64	15.36
2	0.05	0.10	0.02	49.80	159.84	14.38	4.79	6.69	11.33	15.99
3	0.05	0.24	0.04	35.10	115.71	16.51	5.60	7.36	11.94	16.82
4	0.06	0.52	0.09	28.15	94.48	19.49	6.52	8.02	12.51	17.76
High	0.07	1.69	0.31	26.18	84.46	24.29	8.16	9.37	13.38	19.18

Panel B: CVTURN30										
	CVVOL	CVTURN	BETA	SIZE	MOM (%)	Alpha (%)	BM (%)	VOL	TURN (%)	VSHOCK
Low	0.37	0.34	1.02	345.19	-0.20	-0.22	0.09	1.88	0.72	-0.12
2	0.56	0.53	1.05	279.87	-0.14	-0.15	0.07	1.07	0.33	-0.14
3	0.76	0.71	1.03	168.05	-0.09	-0.10	0.07	0.53	0.24	-0.08
4	1.04	0.98	1.00	117.79	0.01	0.00	0.07	0.34	0.16	0.02
High	1.71	1.63	0.94	106.18	0.10	0.08	0.06	0.16	0.06	0.10
	BA	ILLIQ	RILLIQ	NEW	ACT	MAX (%)	IVOL (%)	RVOL (%)	VAR95 (%)	ES95 (%)
Low	0.04	0.04	0.01	49.20	151.09	13.05	4.25	6.21	10.78	15.58
2	0.05	0.10	0.02	56.26	177.83	14.48	4.79	6.73	11.31	15.94
3	0.05	0.24	0.04	36.83	119.65	16.45	5.56	7.32	11.83	16.68
4	0.06	0.53	0.10	28.21	95.03	19.25	6.44	7.95	12.51	17.79
High	0.07	1.79	0.33	23.60	78.25	22.83	7.83	9.07	13.34	19.16

five portfolios. The average bid-ask spread almost doubles from 0.04 to 0.07 between the bottom and top quintiles. The two illiquidity measures – Amihud and realized Amihud – also increase from 0.04 and 0.01 for the lowest variation of trading activity cryptocurrencies to 1.69 to 0.31 for the highest. These results are consistent with the observation that the level of trading activity tends to increase with asset liquidity. For instance, the average trading volume in the first quintile is \$1.93 million compared to just \$0.20 million in the fifth quintile. Considering the prior findings in the literature and the observed liquidity patterns across quintiles, one may attribute the profitability of the CVVOL strategy to

limits-to-arbitrage.

Finally, several risk measures – idiosyncratic and realized volatility as well as value-at-risk and expected shortfall – tend to increase with the coefficient of variation. In equity markets, [Ang et al. \(2006\)](#) document a negative relation between idiosyncratic risk and expected stock returns. In cryptocurrency markets, however, the aforementioned risk measures are found to positively predict future returns ([Zhang and Li, 2020](#); [Zhang et al., 2021](#)). This likely limits the possibility to provide a risk-based explanation of the volume variability effect.

In Section 3, we demonstrate that the effect of the variation of trading activity remains significant (although it diminishes in some cases) after accounting for other cryptocurrency characteristics in conditional bivariate portfolio sorts. To better understand the effect, we construct an arbitrage index that captures different dimensions of arbitrage costs from various proxies. The independently double-sorted portfolios based on the arbitrage index and the coefficient of variation of volume suggest costly arbitrage as an appropriate explanation of the observed effect.

2.6 Transition matrix

In this section, we compute the k -day-ahead portfolio transition matrix to evaluate the cross-sectional persistence of the coefficient of variation of volume.⁴ We calculate the average probability that a cryptocurrency in a specific quintile will be in another quintile k days from today. For a better illustration of portfolio persistence, we examine the transition probabilities at three lags of $k = 1, 14, 30$ days.

Table 5 reports the transition probabilities for quintile portfolios sorted on the volume variation measures estimated over the 7-day (Panel A) and 30-day (Panel B) rolling windows. If the variations in trading volume are random, the probabilities in the matrix should be approximately 20%. However, Panel A indicates that the lowest quintile cryptocurrencies have a probability of 78.9% to be in the same quintile the next day. This probability decreases monotonically to 42.2% and 37.9 % as we move from a 1-day lag to longer periods of 14 and 30 days. Interestingly, the highest quintile cryptocurrencies

⁴The results of the coefficient of variation of turnover are virtually the same and are available upon request.

Table 5. Transition matrix for CVVOL portfolios.

This table reports transition probabilities for quintile portfolios formed daily based on the coefficient of dollar trading volume between September 1, 2017, and September 1, 2022. The table shows results based on the coefficient of variation measures calculated over the 7-day (Panel A) or 30-day (Panel B) rolling windows. For each measure, we compute transition probabilities at the lag of $k = 1, 14, 30$ days. Specifically, we form quintile portfolios of cryptocurrencies based on the variation of the trading volume on the day t . We then form quintile portfolios of cryptocurrencies based on the variation of the trading volume on the day $t + k$. For each quintile formed on the day t , we compute the percentage of cryptocurrencies that belong to each quintile formed on the day $t + k$. For each horizon k , the table reports the time-series averages of these transition probabilities. The rows correspond to quintiles formed at time t , and the columns correspond to quintiles formed at time $t + k$.

Panel A: CVVOL7															
	1 day					14 days					30 days				
	Low	2	3	4	High	Low	2	3	4	High	Low	2	3	4	High
Low	78.9	15.3	2.6	1.0	0.8	42.2	23.3	14.7	9.6	5.9	37.9	22.4	15.4	10.8	6.4
2	15.4	61.8	17.7	2.7	1.3	23.4	27.0	22.2	15.3	8.1	22.9	25.4	21.2	15.4	8.7
3	2.9	18.0	58.9	16.7	2.1	15.1	21.9	24.0	20.7	13.4	15.5	21.5	22.6	19.6	12.7
4	1.0	2.8	17.5	63.2	13.3	10.0	15.7	20.8	25.0	22.0	10.9	15.4	19.7	22.8	21.2
High	0.5	1.1	2.0	14.3	78.8	5.9	8.7	13.4	22.7	40.6	7.0	9.4	13.1	21.1	37.1

Panel B: CVVOL30															
	1 day					14 days					30 days				
	Low	2	3	4	High	Low	2	3	4	High	Low	2	3	4	High
Low	92.7	5.6	0.3	0.1	0.1	61.3	21.0	8.2	3.2	2.3	46.6	22.7	13.0	7.0	4.4
2	5.8	85.6	6.8	0.4	0.2	23.5	38.9	22.0	8.0	3.7	24.9	27.2	22.1	12.5	7.4
3	0.3	7.0	84.2	6.5	0.6	8.0	23.8	34.3	21.6	7.1	12.9	22.4	23.9	20.5	12.1
4	0.1	0.5	6.9	85.1	5.2	3.4	9.7	23.4	37.4	19.6	7.5	14.7	20.5	25.8	21.3
High	0.1	0.2	0.4	5.7	90.4	1.3	3.4	7.8	22.4	56.9	4.2	7.7	13.1	23.2	39.8

remain in the same quintile after 1, 14, and 30 days with similar probabilities of 78.8%, 40.6%, and 37.1%. Panel B reports results for the coefficient of variation measure using a 30-day rolling window. It demonstrates that the transition probabilities of staying in the lowest (highest) quintile become even higher and are monotonically decreasing from 92.7% (90.4%) to 61.3% (56.9%) and 46.6% (39.8%) when we look at the lags from 1 day to 14 and 30 days. In general, cryptocurrencies belonging to the portfolio with the lowest (highest) coefficient of variation tend to stay in the same quintile and have the highest chances to move to the nearest quintile. This suggests that the coefficient of variation of volume is a highly persistent metric. Nevertheless, one can still speculate that transaction costs can limit the significant abnormal returns of the zero-cost strategy, which is rebalanced on a daily basis. We investigate this possibility in robustness checks.

3 Potential explanations

3.1 Market beta, size, past performance, and network-to-market ratio

In this section, we examine the relation between variations in trading activity and expected returns after controlling for the well-known characteristics used to explain cryptocurrency prices: market beta, size, past performance, network-to-market premium, and alpha. Our goal is to disentangle the effect of trading activity variability from common risks. To do so, we construct bivariate portfolios by sorting cryptocurrencies into three unequal-sized groups (30% lowest, 40% middle, 30% highest) using the control variable and then creating quintiles within each third based on the coefficient of variation measures. By construction, the first (fifth) quintile consists of cryptocurrencies with the lowest (highest) variation of the trading volume in each control group. We then calculate the average return of a particular quintile across three groups of a control variable.

Table 6 reports value- and equal-weighted portfolio average returns across the three control groups. Controlling for size, Panel A (Panel B) shows that the average return spread between the extreme CVVOL7 (CVVOL30) quintiles is 17.15% (17.56%) with a significantly positive t-statistic of 7.60 (7.96). In both cases, the return difference also generates a significantly large and positive three-factor alpha of 8.04% and 8.25% per month. Thus, market capitalization cannot explain the abnormal returns of the strategy exploiting the variations in trading activity. Controlling past performance, the results indicate that the cumulative return over the past seven or thirty days cannot explain the return difference of quintiles with the lowest and highest coefficients of variation. Additionally, conditional alpha and beta fail to substantially diminish the average return spread. Finally, we control the network-to-market ratio of individual cryptocurrencies. As shown in the previous section, the network-to-market measure monotonically decreases across portfolios and, hence, the variation of trading volume can be conceived as a proxy for the network effects. However, Table 6 reports significantly large returns for both value- and equal-weighted portfolios when controlling for the network-to-market ratio.

In sum, the comparison between the results of univariate sorts in Table 2 and bivariate sorts in Table 6 indicate minimal changes in key patterns. Even after controlling for the popular cryptocurrency characteristics – market alpha and beta, past momentum, the size

Table 6. CVVOL portfolios after controlling for BETA, SIZE, MOM, ALPHA, BM.

This table reports double-sorted, value- and equal-weighted portfolios formed daily based on the coefficient of variation of volume after controlling for market beta (BETA), market capitalization (SIZE), past 7- and 30-day momentum (MOM7 and MOM30), alpha (ALPHA), and the book-to-market ratio (BM) between September 1, 2017, and September 1, 2022. For each case, we first sort the cryptocurrencies into three unequal-sized groups (30% lowest, 40% middle, 30% highest) using the control variable, then within each third, we sort cryptocurrencies into quintile portfolios based on the coefficient of variation measures calculated over the 7-day (Panel A) or 30-day (Panel B) rolling windows. The table presents average returns across the three control group to produce quintile portfolios with dispersion in volume variations but with similar levels of the control variable. It also reports the mean and alpha of the difference in average returns between the lowest and highest variation of volume portfolios. Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the difference. We report monthly mean and alpha statistics.

	Value-Weighted Portfolios						Equal-Weighted Portfolios					
	BETA	SIZE	MOM7	MOM30	ALPHA	BM	BETA	SIZE	MOM7	MOM30	ALPHA	BM
Panel A: CVVOL7												
Low	-0.96	-0.27	-1.63	0.48	-1.39	-0.79	-0.95	-0.32	-2.09	-1.05	-1.87	-0.84
2	-1.63	-2.51	-2.37	-2.70	-0.23	-1.77	-3.55	-3.36	-4.07	-2.84	-2.69	-2.19
3	-4.66	-6.34	-4.49	-2.91	-4.24	-4.24	-5.77	-5.95	-5.45	-5.24	-4.51	-4.68
4	-6.94	-11.47	-9.48	-8.36	-10.23	-5.02	-9.57	-10.03	-11.40	-9.36	-10.22	-9.11
High	-19.51	-17.41	-14.76	-16.50	-18.21	-17.58	-19.17	-16.36	-15.98	-19.26	-19.13	-18.97
LMH	18.55	17.15	13.13	16.98	16.82	16.79	18.23	16.04	13.89	18.21	17.26	18.13
t-stat	7.49	7.60	5.05	6.50	6.47	7.23	9.04	8.18	6.50	9.12	9.01	9.43
alpha	21.05	17.48	14.97	18.81	18.81	18.37	19.13	15.93	13.79	18.46	17.77	18.65
t-stat	8.42	8.04	5.92	7.36	7.40	7.89	9.67	8.31	6.69	9.30	9.32	9.92
Panel B: CVVOL30												
Low	-0.97	-0.36	-0.30	-0.36	0.94	-1.02	-2.37	-1.43	-2.38	-3.14	-1.68	-1.46
2	-0.17	-4.45	-2.64	-2.96	-4.37	-3.05	-3.46	-5.10	-4.29	-4.49	-5.54	-3.26
3	-7.93	-6.10	-5.36	-3.63	-6.31	-4.20	-7.24	-4.64	-5.52	-5.28	-5.59	-6.12
4	-8.78	-9.84	-10.76	-8.43	-7.39	-8.20	-8.25	-9.35	-11.14	-10.16	-9.24	-7.36
High	-14.97	-17.93	-13.79	-11.86	-13.58	-14.12	-17.81	-15.81	-15.92	-14.83	-16.43	-18.24
LMH	14.00	17.56	13.48	11.49	14.52	13.10	15.44	14.37	13.54	11.69	14.76	16.78
t-stat	5.22	7.96	4.77	4.03	5.02	5.57	7.79	7.51	6.57	6.06	7.04	8.19
alpha	17.16	18.55	16.53	13.60	16.63	16.20	16.73	15.24	14.75	12.45	15.61	18.44
t-stat	6.22	8.25	5.90	4.99	6.24	7.20	8.19	7.91	7.14	6.45	7.42	9.02

and value factors – there is a strong negative relation between the variation in trading activity and future returns.

3.2 Volume, turnover and liquidity measures

We now investigate the connection between variations in trading activity and volume-related cryptocurrency characteristics. First, we construct dependent bivariate portfolios controlling for the level of dollar trading volume and turnover. The reason is that the coefficient of variation measures may be highly correlated with the level of trading activity, which can explain our results. Second, fluctuations in trading volume may be driven by abnormally high or low trades relative to the historical data. Thus, we capture this channel

Table 7. CVVOL portfolios after controlling for VOL, TURN, VSHOCK, BA, ILLIQ, RILLIQ.

This table reports double-sorted, value- and equal-weighted portfolios formed daily based on the coefficient of variation of volume after controlling for volume (VOL), turnover (TURN), volume shocks (VSHOCK), bid-ask spread (BA), Amihud and realized Amihud illiquidity measures (ILLIQ and RILLIQ) between September 1, 2017, and September 1, 2022. For each case, we first sort the cryptocurrencies into three unequal-sized groups (30% lowest, 40% middle, 30% highest) using the control variable, then within each third, we sort cryptocurrencies into quintile portfolios based on the coefficient of variation measures calculated over the 7-day (Panel A) or 30-day (Panel B) rolling windows. The table presents average returns across the three control group to produce quintile portfolios with dispersion in volume variations but with similar levels of the control variable. It also reports the mean and alpha of the difference in average returns between the lowest and highest variation of volume portfolios. Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the difference. We report monthly mean and alpha statistics.

	Value-Weighted Portfolios						Equal-Weighted Portfolios					
	VOL	TURN	VSHOCK	BA	ILLIQ	RILLIQ	VOL	TURN	VSHOCK	BA	ILLIQ	RILLIQ
Panel A: CVVOL7												
Low	-1.83	-1.48	-1.13	-3.00	-1.33	-1.39	-1.24	-1.14	-0.78	-2.30	-1.24	-0.85
2	-1.08	-4.27	-1.46	-1.39	-4.15	-4.26	-3.15	-3.53	-3.47	-3.87	-3.84	-4.05
3	-4.55	-5.85	-4.00	-3.53	-6.08	-6.23	-6.40	-7.29	-5.09	-5.21	-6.08	-6.82
4	-11.15	-6.55	-9.74	-10.22	-10.80	-10.46	-10.36	-7.07	-9.16	-8.90	-11.14	-10.58
High	-16.26	-15.79	-14.08	-17.66	-16.20	-16.67	-17.20	-14.11	-13.21	-17.89	-15.66	-15.67
LMH	14.43	14.31	12.95	14.66	14.87	15.29	15.96	12.97	12.43	15.59	14.41	14.82
t-stat	6.05	5.77	5.43	5.75	5.80	5.80	8.13	6.30	6.89	7.43	7.02	7.03
alpha	15.94	16.41	14.91	15.61	16.16	16.33	15.88	13.63	12.84	15.17	13.77	14.06
t-stat	6.49	6.43	6.34	5.99	6.05	5.96	8.55	6.74	7.07	7.43	6.90	6.99
Panel B: CVVOL30												
Low	-3.02	-1.58	-0.58	-0.87	-2.35	-2.55	-2.73	-3.58	-1.47	-2.95	-2.73	-2.82
2	-4.68	-4.74	-3.12	-3.38	-4.20	-5.19	-4.26	-4.65	-5.11	-5.22	-4.17	-4.33
3	-4.12	-4.08	-6.00	-5.67	-6.64	-5.84	-7.06	-6.12	-5.03	-6.23	-7.61	-7.11
4	-11.33	-7.45	-10.85	-11.51	-11.48	-11.13	-9.36	-8.37	-10.04	-9.44	-9.99	-10.23
High	-12.80	-11.79	-8.92	-14.00	-12.36	-12.28	-15.42	-9.97	-9.89	-15.22	-13.69	-13.73
LMH	9.79	10.21	8.34	13.13	10.01	9.73	12.68	6.38	8.42	12.27	10.95	10.91
t-stat	4.08	4.13	3.23	4.99	3.82	3.75	6.45	3.47	4.82	6.43	5.42	5.47
alpha	12.60	13.25	11.07	15.05	12.21	11.87	13.37	7.51	9.13	13.05	11.56	11.37
t-stat	4.91	5.10	4.53	5.90	4.37	4.33	6.53	4.03	5.26	6.74	5.53	5.56

by defining volume shocks as the log deviation of trading volume from its trend. Third, we control for liquidity effects by using the bid-ask spread, Amihud and realized Amihud illiquidity measures.

Table 7 reports the average returns of the CVVOL portfolios. It shows that the differences between the low and high CVVOL quintiles remain economically and statistically significant, ranging from 12.95% (8.34%) per month with a t-statistic of 5.43 (3.23) for volume shocks to 15.29% (13.13%) per month with a t-statistics of 5.80 (4.99) for realized Amihud (bid-ask spread) in case of the value-weighted CVVOL7 (CVVOL30) strategy. The corresponding values for the equal-weighted return differences are generally higher. These results demonstrate that volume-related characteristics cannot explain the low returns to

high CVVOL cryptocurrencies, through they partially capture the observed effect.

3.3 Lottery demand and extreme returns

As shown in Table 4, as we move from the lowest to the highest variation in trading activity quintiles, the maximum daily return characteristic is monotonically increasing. Thus, cryptocurrencies with the highest variation in volume (turnover) tend to generate lottery-like payoffs. If investors have a preference for such assets, they would be willing to pay higher prices, leading to lower future returns. Thus, lottery demand among investors would explain the negative relation between CVVOL (CVTURN) and expected returns. Table 8 tests this hypothesis by controlling for the maximum daily return during the past 7 and 30 days in double-sorted portfolios. When controlling for maximum daily returns, the raw return and alpha differences based on the CVVOL30 measure become smaller relative to the CVVOL7 case. However, they remain economically large and statistically significant at all conventional levels.

Furthermore, Table 4 demonstrates striking monotonic patterns in idiosyncratic and realized volatility (IVOL and RVOL) across quintiles. Thus, it is plausible that IVOL and RVOL are proxying for the effect of trading activity variability. Table 8 explore this possibility. There is a stronger effect of controlling for idiosyncratic risk and realized volatility measures in value-weighted portfolios. The average raw and risk adjusted difference returns tend to decrease more in the case of the coefficient of variation estimated over the past 30 days. For instance, the value-weighted CVVOL30 (CVVOL7) difference returns controlled by IVOL, RVOL7, and RVOL30 have the mean of 7.28%, 10.33%, and 10.16% (12.86%, 14.12%, and 13.61%) per month with t-statistics of 2.74, 3.74, and 3.87 (4.74, 5.74, and 5.11), respectively.

By construction, maximum daily returns, idiosyncratic and realized volatility measures are highly correlated and, hence, it is not surprising that all of them simultaneously have a strong effect. This evidence coupled with liquidity results further suggests limits-to-arbitrage as a likely explanation of our findings. Indeed, idiosyncratic risk is the primary arbitrage cost according to the extant literature. Intuitively, idiosyncratic risk delays mispricing because undiversified arbitrageurs take small positions in high idiosyncratic risk assets. Therefore, it is worthwhile verifying that the observed negative relation between

Table 8. CVVOL portfolios after controlling for MAX, IVOL, RVOL.

This table reports double-sorted, value- and equal-weighted portfolios formed daily based on the coefficient of variation of volume after controlling for a maximum daily return over the past 7 and 30 days (MAX7 and MAX30), idiosyncratic volatility (IVOL), realized volatility over the past 7 and 30 days (RVOL7 and RVOL30) between September 1, 2017, and September 1, 2022. For each case, we first sort the cryptocurrencies into three unequal-sized groups (30% lowest, 40% middle, 30% highest) using the control variable, then within each third, we sort cryptocurrencies into quintile portfolios based on the coefficient of variation measures calculated over the 7-day (Panel A) or 30-day (Panel B) rolling windows. The table presents average returns across the three control group to produce quintile portfolios with dispersion in volume variations but with similar levels of the control variable. It also reports the mean and alpha of the difference in average returns between the lowest and highest variation of volume portfolios. Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the difference. We report monthly mean and alpha statistics.

	Value-Weighted Portfolios					Equal-Weighted Portfolios				
	MAX7	MAX30	IVOL	RVOL7	RVOL30	MAX7	MAX30	IVOL	RVOL7	RVOL30
Panel A: CVVOL7										
Low	-3.43	-2.27	-3.95	-4.60	-4.04	-2.33	-1.74	-2.34	-2.32	-2.10
2	-2.35	-4.53	-5.86	-3.56	-5.40	-3.89	-5.30	-5.78	-4.40	-4.95
3	-3.16	-3.02	-3.97	-3.51	-3.40	-5.28	-4.71	-6.30	-6.02	-6.01
4	-8.29	-10.61	-12.84	-8.01	-9.35	-9.52	-9.40	-8.84	-7.92	-8.20
High	-17.06	-17.37	-16.81	-18.72	-17.66	-17.22	-17.63	-15.79	-17.31	-17.40
LMH	13.63	15.10	12.86	14.12	13.61	14.89	15.89	13.46	14.98	15.31
t-stat	5.17	5.66	4.74	5.74	5.11	7.24	8.23	6.70	7.64	7.91
alpha	14.71	17.02	14.34	15.25	15.28	14.23	16.28	13.17	14.44	15.02
t-stat	5.66	6.30	5.14	6.02	5.67	7.03	8.39	6.51	7.58	7.78
Panel B: CVVOL30										
Low	-0.62	-1.26	-5.12	-3.39	-4.00	-2.14	-2.43	-5.29	-3.47	-4.25
2	-3.00	-7.87	-7.18	-3.42	-7.32	-4.63	-7.03	-4.97	-4.99	-5.74
3	-5.50	-6.07	-7.50	-5.78	-5.76	-5.08	-6.11	-7.06	-6.05	-6.00
4	-10.04	-7.42	-7.87	-7.39	-7.21	-10.39	-8.65	-8.98	-9.02	-8.04
High	-13.90	-12.32	-12.39	-13.72	-14.16	-16.65	-14.87	-13.25	-15.42	-15.06
LMH	13.28	11.06	7.28	10.33	10.16	14.52	12.44	7.96	11.94	10.81
t-stat	4.74	4.20	2.74	3.74	3.87	7.54	6.29	3.95	6.05	5.44
alpha	16.29	14.61	9.84	13.01	12.74	15.24	13.40	8.92	12.85	11.69
t-stat	5.77	5.68	3.79	4.82	4.99	7.74	6.68	4.47	6.36	5.96

variations in trading activity and expected returns is stronger among cryptocurrencies that are costlier arbitrage.

3.4 Investor interest and downside risk

We now explore the possibility that investor interest can help explain the relationship between the variation of trading activity and cryptocurrency returns. This hypothesis is motivated by the clientele effect in equity returns ([Merton, 1987](#)). In cryptocurrency markets, we do not have information on the holdings of individuals or institutions. We partially explore the clientele hypothesis by using the number of new and active addresses

Table 9. CVVOL portfolios after controlling for NEW, ACT, VAR95, ES95.

This table reports double-sorted, value- and equal-weighted portfolios formed daily based on the coefficient of variation of volume after controlling for a sum of new or active addresses over the past 7 and 30 days (NEW7, NEW30, ACT7, and ACT30), value-at-risk (VAR95), and expected shortfall (ES95) between September 1, 2017, and September 1, 2022. For each case, we first sort the cryptocurrencies into three unequal-sized groups (30% lowest, 40% middle, 30% highest) using the control variable, then within each third, we sort cryptocurrencies into quintile portfolios based on the coefficient of variation measures calculated over the 7-day (Panel A) or 30-day (Panel B) rolling windows. The table presents average returns across the three control group to produce quintile portfolios with dispersion in volume variations but with similar levels of the control variable. It also reports the mean and alpha of the difference in average returns between the lowest and highest variation of volume portfolios. Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the difference. We report monthly mean and alpha statistics.

	Value-Weighted Portfolios						Equal-Weighted Portfolios					
	NEW7	NEW30	ACT7	ACT30	VAR95	ES95	NEW7	NEW30	ACT7	ACT30	VAR95	ES95
Panel A: CVVOL7												
Low	-1.80	-2.01	-2.57	-1.77	-4.26	-4.10	-0.31	-0.50	-0.93	-0.18	-1.74	-1.71
2	-0.83	-2.30	0.22	-1.77	-5.02	-5.97	-2.66	-3.47	-2.12	-3.61	-3.87	-4.10
3	-3.45	-4.11	-4.66	-4.60	-6.38	-4.42	-5.91	-5.52	-6.09	-5.75	-5.08	-5.29
4	-8.27	-9.40	-8.91	-9.79	-9.92	-9.87	-8.15	-8.80	-7.98	-8.38	-8.13	-7.07
High	-18.20	-18.08	-18.09	-18.00	-17.40	-16.75	-19.37	-18.38	-18.79	-18.14	-15.83	-17.20
LMH	16.40	16.07	15.52	16.23	13.15	12.66	19.06	17.88	17.87	17.96	14.09	15.48
t-stat	6.57	6.46	6.06	6.53	5.22	4.67	9.74	9.54	9.00	9.60	6.91	7.59
alpha	17.73	17.96	16.03	17.65	14.53	14.15	19.55	18.56	17.93	18.14	14.79	16.25
t-stat	7.49	6.96	6.45	6.99	5.88	5.27	10.28	9.94	9.33	9.76	7.39	8.06
Panel B: CVVOL30												
Low	-0.57	-1.05	-1.07	-0.56	-3.19	-3.98	-1.94	-2.61	-1.72	-1.50	-2.96	-3.18
2	-2.62	-1.12	-2.77	-2.46	-5.37	-5.64	-3.49	-2.49	-3.79	-4.08	-5.30	-4.80
3	-6.73	-6.51	-6.19	-5.85	-6.14	-6.88	-6.40	-6.60	-6.23	-6.13	-4.67	-5.00
4	-8.45	-9.70	-9.22	-9.64	-10.53	-8.83	-7.72	-8.59	-7.60	-7.95	-7.68	-7.98
High	-15.46	-17.01	-15.32	-17.76	-14.91	-13.67	-17.52	-16.58	-16.95	-16.65	-14.26	-14.27
LMH	14.89	15.97	14.25	17.20	11.72	9.69	15.58	13.97	15.23	15.15	11.29	11.09
t-stat	5.98	6.17	5.51	6.54	4.60	3.42	8.28	7.24	7.97	7.37	5.68	5.59
alpha	16.61	18.46	16.51	19.67	13.69	12.47	16.72	15.21	16.02	16.15	12.41	12.78
t-stat	6.74	7.09	6.38	7.33	5.48	4.60	8.67	7.72	8.24	7.73	6.15	6.40

as a proxy for the investor following. As shown in Section 2.5, the average number of new and active addresses is decreasing in the variability of trading activity. In unreported results, we document, at best, a weakly positive relation between new (active) addresses and expected returns. Consequently, a larger user network could be associated with lower abnormal variation in trading activity, which would explain the negative relation of the latter with future returns. Table 9 presents double-sorted portfolios when controlling for the average number of new and active addresses over the past 7- or 30-day period. The results are not materially different from those provided for the univariate sorts.

We also test whether downside risk can explain the negative relation between the variation of trading activity and expected returns. [Zhang et al. \(2021\)](#) demonstrate that cryp-

to currencies with higher downside risk have higher returns. As shown in Section 2.5, the highest (lowest) CVVOL and CVTURN assets tend to have higher value-at-risk and expected shortfall statistics. Thus, if downside risk was a plausible explanation of the documented effect, we should have observed the opposite association, making a downside risk premium an unlikely explanation. Table 9 shows that controlling for VaR95 and ES95 only slightly reduces the average return and alpha of the difference returns, having a stronger effect on the CVVOL30 portfolio.

3.5 Cryptocurrency-level cross-sectional regressions

The single- and double-sorted portfolios demonstrate a significant negative relation between variations in trading activity and expected returns, which cannot be subsumed by other cryptocurrency characteristics. We now examine this cross-sectional relation using the standard Fama and MacBeth (1973) regressions.⁵ Each day, we estimate a cross-sectional regression of one-day-ahead cryptocurrency returns on the coefficient of variation of volume and control variables, one at a time. We then compute the time-series averages of the intercept and slope coefficients, their Newey and West (1986) adjusted t-statistics, and average R-squared statistics.

Table 10 reports the results. The cross-sectional analysis confirms a statistically significant negative relationship between variations in trading activity and future cryptocurrency returns. In the univariate regression of one-day-ahead returns on CVVOL, the average slope coefficient is -0.0052 (-0.0028) with a t-statistic of -9.28 (-7.39) for the case of the 7-day (30-day) rolling window. When we add control variables one at a time, the CVVOL coefficient tends to decrease but remains significant in all cases. Consistent with dependant double-sorted portfolios, the economic magnitude and statistical significance of the CVVOL slope weaken the most when controlling for volume shocks, past maximum daily return, liquidity and volatility measures. Although the CVVOL effect becomes weaker, it remains statistically significant at all conventional confidence levels. We further multiply the CVVOL estimates by the spread in median CVVOL characteristics between the lowest and highest CVVOL quintiles to proxy for the CVVOL risk premium. The last row in

⁵The main advantage of the portfolio-level analysis is a non-parametric form, that is, it does not impose any functional dependence of returns on characteristics. The aggregation of cryptocurrency returns in portfolios, however, ignores a large amount of information in the cross-section. For this reason, we also look at the cross-sectional regressions of Fama and MacBeth (1973) that account for this information.

Table 10. Cryptocurrency-level cross-sectional regressions.

This table reports daily [Fama and MacBeth \(1973\)](#) results for cross-sectional regressions between September 1, 2017, and September 1, 2022. Each day, we estimate a cross-sectional regression of one-day-ahead cryptocurrency returns on the coefficient of variation of volume and control variables, one at a time. We then report the time-series averages of the estimates, their [Newey and West \(1986\)](#) adjusted t-statistics (cells in the grey font), and average R-squared statistics. The last row in each panel reports the monthly CVVOL risk premium, which is calculated as the product of the CVVOL coefficient and the spread in median CVVOL characteristics between the lowest and highest CVVOL quintiles. The table shows results based on the coefficient of variation measures calculated over the 7-day (Panel A) or 30-day (Panel B) rolling windows. The control variables include CAPM beta (BETA); market capitalization (SIZE); past 7- and 30-day momentum (MOM7 and MOM30), network-to-market ratio (BM); volume (VOL); volume shocks (VSHOCK); bid-ask spread (Bid-Ask); Amihud and realized illiquidity measures (ILLIQ and RILLIQ); a maximum daily return over the past 7 and 30 days (MAX7 and MAX30), idiosyncratic volatility (IVOL), realized volatility over the past 7 and 30 days (RVOL7 and RVOL30), value-at-risk (VAR95), and expected shortfall (ES95).

Panel A: 7-day look-back period															
	CVVOL7	BETA	SIZE	MOM7	BM	VOL	VSHOCK	BA	ILLIQ	RILLIQ	MAX7	IVOL	RVOL7	Var95	ES95
Intercept	0.0008	0.0012	0.0007	-0.0002	0.0007	0.0007	0.0001	0.0034	0.0008	0.0008	0.0020	0.0032	0.0033	0.0046	0.0041
CVVOL7	0.61	0.97	0.55	-0.18	0.52	0.55	0.05	2.45	0.61	0.60	1.53	2.45	2.47	3.37	2.92
	-0.0052	-0.0052	-0.0052	-0.0045	-0.0052	-0.0051	-0.0037	-0.0031	-0.0049	-0.0049	-0.0032	-0.0033	-0.0031	-0.0037	-0.0040
	-9.28	-9.85	-9.45	-8.88	-9.52	-9.04	-6.91	-5.19	-8.74	-8.73	-5.60	-5.83	-5.44	-6.31	-6.80
CONTROL	-0.0005	0.0154	-0.0346	0.0083	0.0083	0.0045	-0.0003	-0.0660	0.0012	0.0110	-0.0241	-0.0534	-0.0504	-0.0376	-0.0228
R^2_{adj}	-0.62	2.25	-2.14	0.38	2.67	2.67	-1.00	-8.21	0.36	0.54	-5.03	-5.12	-5.68	-5.71	-4.97
	1.01%	3.83%	0.19%	4.57%	0.63%	0.27%	1.49%	5.87%	5.16%	5.17%	5.26%	4.99%	5.64%	5.01%	4.60%
RP(CVVOL7)	-15.37	-15.44	-15.59	-13.31	-15.57	-15.25	-10.91	-9.26	-14.63	-14.53	-9.51	-9.99	-9.29	-11.08	-12.00
Panel B: 30-day look-back period															
	CVVOL30	BETA	SIZE	MOM30	BM	VOL	VSHOCK	BA	ILLIQ	RILLIQ	MAX30	IVOL	RVOL30	Var95	ES95
Intercept	0.0002	0.0005	0.0002	-0.0001	0.0001	0.0000	-0.0006	0.0031	-0.0001	-0.0001	0.0017	0.0023	0.0035	0.0039	0.0034
CVVOL30	0.13	0.38	0.14	-0.09	0.08	-0.02	-0.46	2.27	-0.06	-0.05	1.35	1.81	2.64	2.85	2.44
	-0.0028	-0.0028	-0.0030	-0.0026	-0.0029	-0.0027	-0.0017	-0.0016	-0.0024	-0.0024	-0.0014	-0.0014	-0.0013	-0.0016	-0.0018
	-7.39	-7.36	-7.34	-6.56	-7.29	-6.96	-4.76	-4.11	-6.21	-6.19	-3.33	-2.45	-2.57	-3.92	-4.38
CONTROL	-0.0004	0.0157	-0.0542	0.0232	0.0232	0.0051	-0.0003	-0.0680	0.0015	0.0112	-0.0144	-0.0583	-0.0612	-0.0383	-0.0235
R^2_{adj}	-0.51	2.52	-1.44	1.31	3.19	3.19	-1.22	-8.43	0.34	0.46	-4.50	-5.41	-5.27	-5.85	-5.16
	0.82%	3.67%	-0.03%	3.56%	0.41%	0.07%	1.29%	5.70%	5.03%	5.04%	3.84%	4.80%	4.93%	4.82%	4.38%
RP(CVVOL30)	-11.42	-11.29	-11.96	-10.58	-11.86	-10.80	-6.81	-6.66	-9.60	-9.59	-5.56	-5.46	-5.30	-6.35	-7.08

Table 10 reports the corresponding values on a monthly basis.⁶ For instance, the CVVOL7 slope coefficient from the univariate regression yields a monthly risk premium of 15.37%, which compares well with the mean value-weighted return of 16.36% reported in Table 2. In general, the reported estimates are comparable to the monthly statistics of single- and double-sorted differences in Table 2 and Table 6 through Table 9.

3.6 Costly arbitrage

The cryptocurrencies with a higher variation of trading activity tend to experience strong recent performance measured by a cumulative raw return, risk-adjusted alpha, or a maximum daily return. Naturally, this may attract retail and institutional investors with a preference for assets with strong momentum and lottery-like payoffs. Thus, they would overprice the cryptocurrencies with the abnormal variations of trading volume and be willing to pay higher prices. As other market participants look for inefficiencies to identify potential arbitrage opportunities, one may argue that competitive players should detect and benefit from this mispricing. However, as shown previously in the literature (Acharya et al., 2013), taking advantage of such arbitrage opportunities can be costly to execute. The results based on double-sorted portfolios cannot provide a complete explanation of the effect of trading activity variation. However, the evidence related to illiquidity and idiosyncratic risk suggests that limits-to-arbitrage indeed may explain the mispricing of cryptocurrencies with the higher variation of trading activity.

We now formally test the hypothesis of costly arbitrage. Specifically, we construct an arbitrage index that captures several dimensions of limits-to-arbitrage through proxies of idiosyncratic risk, transaction costs, and illiquidity of cryptocurrencies. To construct the arbitrage index, we first utilize (1) idiosyncratic volatility, (2) bid-ask spread, (3) Amihud illiquidity, and (4) realized Amihud illiquidity measures to sort cryptocurrencies into five quintiles. For each of the four measures, we then assign a quintile rank to each cryptocurrency and construct the arbitrage index as a sum of the four ranks. By construction, for each cryptocurrency, the arbitrage index ranges from 4 to 20. We then independently sort the cryptocurrencies into three unequal-sized groups (30% lowest, 40% middle, 30%

⁶As shown in Table 4, spread in median CVVOL30 characteristics between the lowest and highest CVVOL30 quintiles is around 1.347. In the CVVOL07 case, the spread is 0.995. One can use these values to compute the risk premiums reported in Table 10.

Table 11. Costly arbitrage.

This table reports independently double-sorted, value-weighted (Panel A) and equal-weighted (Panel B) portfolios formed daily based on the arbitrage index and the coefficient of variation of dollar trading volume between September 1, 2017, and September 1, 2022. For the variation of the trading volume, we consider 7-, 14-, 21-, and 30-day rolling window estimates. For the arbitrage index, each day we sort cryptocurrencies into quintiles according to their idiosyncratic volatility, bid-ask spread, Amihud and realized Amihud illiquidity measures. Based on these rankings, we assign a quintile rank to each cryptocurrency and construct the arbitrage index as a sum of the four ranks. We then independently sort the cryptocurrencies into three unequal-sized groups (30% lowest, 40% middle, 30% highest) based on the arbitrage index and quintile portfolios based on the variation of the trading volume. The table presents average returns of 5×3 portfolios. It also reports the mean and alpha of the difference in average returns between the lowest and highest variation of the trading volume portfolios within each arbitrage index group. Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the difference. We report monthly mean and alpha statistics.

	CVVOL7			CVVOL14			CVVOL21			CVVOL30		
	AI1	AI2	AI3	AI1	AI2	AI3	AI1	AI2	AI3	AI1	AI2	AI3
Panel A: Value-Weighted Portfolios												
Low	0.70	-2.15	1.49	0.17	-3.73	-3.04	0.42	-3.05	0.10	1.69	-2.50	0.77
2	1.86	-5.09	-3.67	1.18	-5.43	-1.64	2.35	-6.85	-9.35	-1.55	-8.63	-8.31
3	-1.36	-5.39	-4.24	-2.04	-6.86	-7.35	-1.54	-8.61	-9.33	-1.52	-4.90	-15.54
4	-1.95	-10.75	-14.29	-1.57	-2.79	-16.21	-0.54	-4.51	-11.45	-3.65	-2.67	-14.47
High	-2.09	-9.36	-28.34	-1.36	-10.82	-27.69	-2.37	-7.58	-29.37	-1.91	-7.57	-25.41
LMH	2.79	7.21	29.84	1.53	7.09	24.65	2.79	4.53	29.47	3.61	5.06	26.18
t-stat	0.76	1.68	3.80	0.43	1.78	3.31	0.80	1.03	4.00	1.06	1.18	3.16
alpha	3.66	10.02	26.69	2.98	8.00	25.40	3.21	4.47	31.93	3.53	6.45	25.97
t-stat	1.01	2.24	3.47	0.81	2.08	3.25	0.87	1.02	4.37	1.03	1.49	3.22
Panel B: Equal-Weighted Portfolios												
Low	0.29	1.05	1.16	-0.48	-3.25	-4.84	-0.70	-2.84	-2.27	0.16	-1.23	-3.94
2	-1.33	-3.58	-7.15	-1.58	-3.05	-6.52	-1.73	-2.98	-8.95	-2.67	-5.97	-6.45
3	-2.11	-4.00	-5.75	-0.54	-4.09	-6.48	-1.85	-4.04	-7.57	-2.50	-3.05	-13.63
4	-1.55	-6.80	-16.11	-1.48	-3.67	-13.14	-2.35	-4.62	-10.58	-2.07	-4.73	-11.95
High	-2.37	-11.03	-25.56	-1.37	-9.82	-27.01	-1.39	-8.05	-28.08	-1.32	-5.91	-25.72
LMH	2.66	12.08	26.72	0.88	6.57	22.17	0.68	5.21	25.81	1.49	4.69	21.78
t-stat	0.77	4.33	4.15	0.28	2.77	3.77	0.22	1.88	4.34	0.48	1.70	3.24
alpha	1.50	13.34	23.73	0.21	6.97	22.93	-1.19	4.31	28.53	-0.58	4.67	22.33
t-stat	0.45	4.73	3.83	0.06	2.88	3.76	-0.36	1.55	4.71	-0.19	1.65	3.39

highest) based on the arbitrage index and quintile portfolios based on the variation of the trading volume.

Table 11 presents average returns of independently double-sorted portfolios. Controlling for the level of arbitrage costs, we document that the volume variation strategy formed on cryptocurrencies within the first (AI1) or second group (AI2) of the arbitrage index fails to deliver significant abnormal returns. That is, cryptocurrencies with lower costs to arbitrage are not prone to mispricing. In contrast, we observe a pronounced effect of volume variation among cryptocurrencies in the group with the highest arbitrage cost (AI3). For

the highest arbitrage index group, the magnitude of the average return to the zero-cost portfolio ranges from 18.24% to 35.08% per month. Thus, we empirically confirm our conjecture that the mispricing of cryptocurrencies with the high variation of trading volume increases with arbitrage costs.

3.7 *Alternative screens*

Throughout our analysis, we implement various single- and double-sorted portfolios by choosing the top 100 cryptocurrencies according to market capitalization on a particular day. The advantage of this approach is that the number of coins in (dependent) quintile sorts remains constant over time, while simultaneously excluding the smallest market-value coins. This section tests the robustness of key findings to alternative portfolio screens. We first limit the investable space to the top 50 cryptocurrencies based on market capitalization. Consistent with [Cong et al. \(2021\)](#) and [Liu et al. \(2022\)](#), we then present sensitivity results when including only digital assets with market values larger than \$1 million. In a final check, we follow [Zhang et al. \(2021\)](#) and exclude cryptocurrencies whose Amihud illiquidity belongs to the top tercile (the least liquid third of coins) on a particular day.

Table 12 presents the results of the unidimensional portfolio analysis for different screens. Several observations are noteworthy. First, the negative relation between trading activity variability and expected returns is robust to alternative filters. For all screens and portfolios considered, the zero-cost strategy generates a positive average return, which is economically and statistically significant. Second, applying size-based filters (either the top 50 or a 1-million screen) tends to have a stronger effect on the equal-weighted results as one observes a bigger reduction in the average difference returns compared to the original analysis. Intuitively, more restrictive size screens eliminate smaller cryptocurrencies that likely belong to the highest coefficient of variation quintile, diminishing the profitability of the long-short investment. Third, for the liquidity filter, the excess returns become even smaller but remain significantly positive in all cases.

Overall, this evidence indicates that cryptocurrency screening based on liquidity has a non-negligible effect on the relationship between trading activity variation and cryptocurrency returns, which is consistent with the limits-to-arbitrage explanation provided before.

Table 12. CVVOL and CVTURN portfolios for alternative screens.

This table reports summary statistics of value- and equal-weighted quintile portfolios formed daily based on the coefficient of variation of dollar trading volume or turnover between September 1, 2017, and September 1, 2022. The table also reports the returns on the long-short portfolio buying and selling quintiles of cryptocurrencies with the lowest and highest variations of the trading volume. Panels A, B, C, and D present the returns on portfolios based on the coefficient of variation measures calculated over the 7-, 14-, 21-, and 30-day rolling windows, respectively. For portfolio formation, each period we restrict our sample to cryptocurrencies that have (1) a market capitalization in the top 50 of the sample (a “top50” column); (2) a market capitalization larger than 1 million dollars (a “>\$1m” column); and (3) Amihud liquidity measure in the two lowest terciles (a “ILLIQ” column). The table presents the average returns of quintile portfolios. It also reports the mean and alpha of the difference in average returns between the lowest and highest variation of the trading volume (turnover) portfolios. Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the long-short portfolio. We report monthly mean and alpha statistics.

	Value-Weighted Portfolios						Equal-Weighted Portfolios					
	CVVOL			CVTURN			CVVOL			CVTURN		
	top50	>\$1m	ILLIQ	top50	>\$1m	ILLIQ	top50	>\$1m	ILLIQ	top50	>\$1m	ILLIQ
Panel A: 7-day look-back period												
Low	-2.82	-1.10	-1.44	-2.83	-2.63	-3.48	-2.56	-0.07	-0.36	-2.60	-1.38	-0.82
2	2.73	2.59	0.54	-0.16	1.95	1.15	0.17	-2.46	-1.20	-0.81	-2.60	-1.73
3	-2.46	-1.75	-3.55	-0.75	-4.22	-2.23	-3.24	-5.52	-4.71	-3.94	-5.23	-4.62
4	-6.09	-14.44	-5.98	-4.08	-9.56	-3.47	-7.57	-7.45	-3.49	-6.59	-8.85	-3.93
High	-16.42	-18.99	-13.52	-15.14	-19.62	-13.69	-18.66	-13.20	-10.84	-17.92	-10.62	-9.54
LMH	13.60	17.89	12.08	12.32	16.99	10.20	16.10	13.13	10.47	15.33	9.24	8.72
t-stat	3.99	4.35	4.18	3.42	4.31	3.05	7.34	7.55	6.43	6.67	5.39	5.54
Panel B: 14-day look-back period												
Low	1.07	0.71	-1.27	-0.04	0.01	-0.37	-0.81	-1.69	-2.29	-2.05	-2.05	-1.67
2	0.30	-0.49	1.00	0.83	-0.39	-0.04	-1.38	-2.63	-0.77	-1.26	-2.99	-3.19
3	-3.80	-7.07	-3.78	-3.81	-7.61	-3.51	-5.03	-3.05	-3.31	-3.02	-3.68	-2.35
4	-7.42	-8.58	-5.87	-6.32	-7.93	-5.62	-6.77	-8.69	-4.84	-8.55	-8.97	-4.94
High	-14.09	-17.76	-13.51	-11.22	-14.20	-9.94	-17.07	-12.76	-9.29	-16.17	-11.15	-8.35
LMH	15.17	18.47	12.24	11.18	14.21	9.56	16.26	11.07	6.99	14.11	9.10	6.69
t-stat	4.22	4.86	3.22	3.03	3.80	2.59	7.23	6.56	4.53	6.28	5.35	4.35
Panel C: 21-day look-back period												
Low	0.22	1.18	-0.42	-0.34	-1.42	-1.41	-0.45	-2.07	-1.89	-1.65	-2.16	-2.28
2	2.54	-0.56	1.34	0.76	-1.83	2.10	-1.81	-2.43	-2.13	-2.29	-3.99	-3.27
3	-3.65	-9.51	-4.48	-1.38	-4.94	-3.18	-4.30	-4.57	-2.32	-2.75	-4.17	-1.82
4	-6.58	-10.65	-6.98	-5.97	-10.08	-7.08	-7.94	-7.34	-5.68	-9.72	-7.85	-5.61
High	-14.40	-15.56	-12.03	-11.59	-13.98	-9.35	-17.02	-12.15	-8.73	-15.10	-10.35	-7.72
LMH	14.62	16.73	11.61	11.26	12.57	7.94	16.57	10.08	6.84	13.45	8.19	5.44
t-stat	3.98	4.48	3.23	3.12	3.37	2.22	6.83	6.77	4.42	5.72	5.52	3.66
Panel D: 30-day look-back period												
Low	1.54	1.33	1.83	-0.90	0.56	2.73	0.84	-1.77	-0.67	0.40	-1.85	-1.44
2	0.60	-3.76	-1.07	0.18	-2.23	-0.87	-1.66	-2.95	-3.46	-3.74	-4.59	-3.64
3	-4.58	-8.69	-6.08	-3.66	-4.44	-5.04	-5.35	-6.86	-3.66	-4.28	-5.36	-4.37
4	-6.85	-9.14	-8.57	-4.32	-11.50	-3.77	-9.09	-5.14	-4.96	-9.29	-7.30	-4.56
High	-15.02	-16.10	-10.93	-11.95	-14.72	-6.54	-15.90	-12.25	-8.06	-14.24	-9.95	-6.80
LMH	16.56	17.43	12.76	11.05	15.29	9.27	16.74	10.48	7.39	14.64	8.10	5.37
t-stat	4.36	4.77	3.81	3.01	3.75	2.44	6.49	6.39	4.86	5.46	4.96	3.71

Market capitalization accounts for a smaller part of the variability of trading activity premium. Nevertheless, none of these screens can fully explain the observed effect.

Table 13. CVVOL and CVTURN portfolios including transaction costs.

This table reports summary statistics of value- and equal-weighted portfolios formed daily based on the coefficient of variation of dollar trading volume (a top panel) or turnover (a bottom panel) between September 1, 2017, and September 1, 2022. The table reports the returns on the long-short portfolio buying and selling quintiles of cryptocurrencies with the lowest and highest variations of the trading volume or turnover calculated over the 7-, 14-, 21-, and 30-day rolling windows. Summary statistics include the mean, standard deviation, Sharpe ratio, alpha (reported on a monthly basis), and skewness (based on the daily portfolio returns). Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the long-short portfolio.

	Value-Weighted Portfolios						Equal-Weighted Portfolios					
	Mean	Std	SR	Skew	alpha	t-stat	Mean	Std	SR	Skew	alpha	t-stat
CVVOL7	8.99	35.88	0.25	-0.31	7.73	2.22	13.19	17.88	0.74	0.12	12.47	6.41
t-stat	2.79						6.76					
CVVOL14	12.92	32.83	0.39	-1.06	13.02	3.25	13.34	17.05	0.78	0.07	13.62	6.98
t-stat	3.47						6.90					
CVVOL21	13.26	31.29	0.42	-1.16	14.02	3.34	14.72	16.48	0.89	0.20	15.00	8.06
t-stat	3.51						8.08					
CVVOL30	14.50	33.13	0.44	-0.39	14.79	3.58	14.10	16.59	0.85	-0.12	14.04	7.12
t-stat	4.02						7.39					
CVTURN7	6.11	33.21	0.18	-0.07	5.12	1.41	10.77	17.30	0.62	0.26	9.88	5.07
t-stat	1.75						5.65					
CVTURN14	7.12	33.32	0.21	-0.24	7.37	1.82	12.36	16.97	0.73	0.11	12.76	6.82
t-stat	1.72						6.51					
CVTURN21	9.40	31.92	0.29	-0.71	11.27	2.79	12.68	16.40	0.77	0.13	13.17	7.08
t-stat	2.48						6.77					
CVTURN30	9.62	32.65	0.29	-0.69	11.13	2.93	12.55	16.80	0.75	-0.02	12.70	6.12
t-stat	2.66						6.11					

3.8 Transaction costs

Although the aim of our article is not to provide a recipe for a real-time trading strategy, it is informative to examine the extent to which portfolio returns would survive transaction costs. One can argue that a combination of a portfolio screen and the lack of persistence in the variation metrics may lead to frequent rebalancing and high turnover. In turn, this could substantially diminish the investment attractiveness of the strategy. To address this concern, we implement the portfolio sorts taking into account transaction costs. The fees vary substantially across different centralized or decentralized exchanges and depend on the trade size, being the lowest for median transactions and the highest for small and large transactions. We apply a simplistic transaction fee of 20 basis points for the long and short sides of the strategy to approximate trading frictions.

Table 13 reports summary statistics of net-of-fees returns of single-sorted long-short

portfolios. Several observations are noteworthy. First, we conclude that the abnormal profitability of the strategy exploiting the variation of trading activity survives transaction costs. Except for a few nuances, the long-short portfolios tend to produce economically and statistically significant average returns and alphas. Second, transaction costs lead to a larger reduction in the average portfolio returns for the volume (turnover) variation estimated over a shorter horizon. Indeed, the value-weighted portfolio based on CVVOL7 (CVTURN7) has an average return of 8.99% (6.11%), whereas the mean return of the CVVOL30 (CVTURN30) portfolio is 14.50% (9.62%). This deterioration in profitability is driven by a larger turnover implied by weaker persistence of CVVOL7 (CVTURN7) compared to CVVOL30 (CVTURN30), as indicated by the transition matrix results in Section 2.6. Third, the equal-weighted strategies experience a more minor reduction in average returns. This happens because the investor pays smaller transaction fees to maintain the equal weights of cryptocurrencies in his portfolio.

4 Conclusion

This paper examines the relationship between the fluctuations in trading activity, proxied by dollar trading volume and turnover, and the cross-sectional differences in expected cryptocurrency returns. We construct the coefficient of variation of volume (turnover) estimated based on various rolling window lengths and investigate whether these trading activity variability measures predict one-day-ahead returns. The univariate portfolio-level analysis shows that crypto assets with a higher variation of trading activity earn lower future returns. The return difference between assets in the lowest and highest coefficient of variation quintiles is economically large, statistically significant, and generates the positive and significant alpha after adjusting for the market, size, and momentum factors of [Liu et al. \(2022\)](#). To better understand the source of this anomaly, we show that cryptocurrencies with the highest coefficient of trading activity variation are smaller, less liquid, more volatile, and prone to higher idiosyncratic and downside risks. These cryptocurrencies also tend to have stronger past momentum and exhibit lottery-like payoffs.

The bivariate portfolio-level analysis and the cross-sectional cryptocurrency-level regressions demonstrate that various cryptocurrency characteristics cannot provide a complete explanation for the negative relationship between volume (turnover) variations and

expected returns. The effect remains highly significant when controlling for market beta, size, past performance, and network premium proxied by the network-to-market ratio. Furthermore, the negative relation survives controls of the level of volume and turnover, volume shocks, liquidity measures, lottery demand, downside risk, and idiosyncratic and realized volatility. Having established the persistent negative relation between variations in trading activity and expected returns, we provide an explanation for this anomaly based on the limits-to-arbitrage theory. In particular, we construct an arbitrage index based on idiosyncratic volatility, bid-ask spread, and (realized) Amihud illiquidity measures. We show that mispricing is more pronounced among cryptocurrencies with higher values of the arbitrage index, that is, those that are costlier to arbitrage. Finally, we demonstrate that the effect of trading activity fluctuations is robust to alternative portfolio screens and survives simplistic transaction costs.

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Appendix for

“Variations in Trading Activity, Costly Arbitrage, and Cryptocurrency Returns”

Abstract

This appendix provides a detailed description of the data construction and additional results not included in the main body of the paper.

A Data cleaning

This appendix presents the procedure used to source, clean, and prepare the data.

A.1 Original sources

1. We download aggregate and exchange-level price and volume data from *CryptoCompare* on a daily frequency, where a day is defined with a start time of 00:00:00 UTC. We set *tryConversion* to “true” and *tsym* to “USD” for the data aggregated across different exchanges. We set *tryConversion* to “false” and *tsym* to “USDT” for the data from individual exchanges.
2. We retrieve the information on blockchain activity including the number of new addresses and active unique addresses from *IntoTheBlock*.

We merge the data from *CryptoCompare/IntoTheBlock* and retain those cryptocurrencies that have all the available information.

A.2 Data pre-processing

We further apply various filters to clean the data.

1. We exclude the observations with a zero trading volume or a zero price.
2. We exclude the observations with the ratio of trading volume to market capitalization above 1 on a particular day. This screens out the data with likely fake volume.
3. Using the classification from *CoinMarketCap*, we exclude digital currencies that are:
 - linked to, backed by, or track the price of gold or any precious metal;
 - wrapped coins (e.g., WBTC);
 - stablecoins including those that are centralized (e.g., USDT, USDC) and algorithmically stabilized (e.g., DAI, UST) for any fiat currencies;
 - centralized exchange-based derivatives.
4. *CryptoCompare* implements additional filters to mitigate the effect of suspicious trading activity.

- Outliers are automatically excluded from the calculation of trading volume and, hence, the volume-weighted scheme. A particular trade is considered an outlier if it deviates significantly either from the median of the set of exchanges or from the previous aggregate price.⁷
- Exchanges are reviewed on a regular basis for each cryptocurrency. Constituent exchanges are excluded if (1) posted prices are too volatile compared to the market average of a given pair; (2) trading has been suspended by the exchange on a given day; (3) verified user or social media report false data provision; or (4) malfunctioning of their public API happens.

B Additional results

This appendix provides a set of additional results not included in the main body of the paper. Tables A1-A4 provide results of double-sorted, value- and equal-weighted portfolios sorted on the coefficient of variation of turnover controlling for various cryptocurrency characteristics. Several observations are noteworthy. First, the key findings are robust to using turnover as a proxy for trading activity. The mean and the alpha of the return difference between the lowest and highest coefficient of variation quintiles remain economically and statistically significant for all control variables regardless of the equal- or value-weighted schemes. Second, the magnitudes of average return differences and their alphas tend to be slightly smaller compared to the volume-based results. In relative terms, volume shocks, liquidity proxies, lottery demand, and volatility measures seem to capture a large fraction of the observed effect of liquidity variability, in line with previous results based on trading volume. Interestingly, the network premium and the level of volume and turnover seem to explain a larger part of the return spread compared, particularly for the CVTURN30 measure. Third, the magnitude of difference returns tends to be larger for equal-weighted portfolios due to a larger effect of smaller cryptocurrencies that receive a larger weight under this scheme. Table A5 further shows that the key findings are robust to the cross-sectional regressions of Fama and MacBeth (1973).

⁷Volume deviations can occur for a number of reasons, such as extremely low liquidity, erroneous data, or incorrect mapping of a cryptocurrency in API.

Table A1. CVTURN portfolios after controlling for BETA, SIZE, MOM, ALPHA, BM.

This table reports double-sorted, value- and equal-weighted portfolios formed daily based on the coefficient of variation of turnover after controlling for market beta (BETA), market capitalization (SIZE), past 7- and 30-day momentum (MOM7 and MOM30), alpha (ALPHA), and the book-to-market ratio (BM) between September 1, 2017, and September 1, 2022. For each case, we first sort the cryptocurrencies into three unequal-sized groups (30% lowest, 40% middle, 30% highest) using the control variable, then within each third, we sort cryptocurrencies into quintile portfolios based on the coefficient of variation measures calculated over the 7-day (Panel A) or 30-day (Panel B) rolling windows. The table presents average returns across the three control group to produce quintile portfolios with dispersion in turnover variations but with similar levels of the control variable. It also reports the mean and alpha of the difference in average returns between the lowest and highest variation of turnover portfolios. Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the difference. We report monthly mean and alpha statistics.

	Value-Weighted Portfolios						Equal-Weighted Portfolios					
	BETA	SIZE	MOM7	MOM30	ALPHA	BM	BETA	SIZE	MOM7	MOM30	ALPHA	BM
Panel A: CVTURN7												
Low	-1.66	-1.80	-3.87	-1.77	-1.73	-3.11	-2.74	-2.12	-2.73	-2.89	-2.78	-2.57
2	-1.39	-2.74	0.36	-0.88	0.38	-0.86	-3.78	-2.92	-1.25	-2.59	-2.67	-1.74
3	-4.78	-4.70	-5.91	-4.77	-4.49	-2.91	-5.29	-5.35	-6.16	-4.71	-4.48	-3.19
4	-9.96	-11.10	-6.36	-8.44	-10.10	-6.13	-9.65	-9.98	-8.56	-9.63	-9.98	-10.17
High	-15.35	-16.08	-17.50	-17.66	-17.36	-16.93	-17.52	-15.63	-18.91	-18.12	-18.53	-18.10
LMH	13.69	14.29	13.63	15.88	15.63	13.83	14.78	13.51	16.18	15.24	15.75	15.53
t-stat	5.08	6.50	5.47	6.14	5.87	6.00	7.42	7.06	8.35	7.57	7.89	8.28
alpha	16.43	14.39	15.47	17.97	17.64	14.72	15.27	13.26	16.27	15.33	15.86	15.41
t-stat	6.24	6.84	6.36	6.96	6.89	6.40	7.74	7.08	8.42	7.66	8.05	8.39
Panel B: CVTURN30												
Low	-3.04	-1.72	-4.54	-2.97	-1.88	-4.43	-2.99	-1.85	-2.66	-3.81	-3.71	-2.97
2	-3.36	-3.83	-1.89	-2.96	-1.70	-1.99	-5.29	-5.34	-4.84	-3.22	-3.63	-3.39
3	-5.10	-6.22	-4.50	-3.15	-4.32	-3.09	-5.49	-5.76	-5.37	-4.75	-5.59	-5.25
4	-10.35	-9.09	-11.63	-7.24	-7.65	-7.55	-10.30	-9.24	-9.85	-9.54	-9.37	-7.58
High	-11.33	-14.84	-12.40	-13.15	-12.18	-11.95	-15.12	-13.99	-15.82	-16.87	-16.21	-17.38
LMH	8.29	13.13	7.86	10.18	10.31	7.52	12.13	12.14	13.17	13.05	12.50	14.41
t-stat	2.77	5.98	2.91	3.48	3.31	2.72	5.76	5.95	6.41	6.16	5.91	6.69
alpha	11.94	13.95	10.16	12.24	12.24	10.18	13.38	12.72	14.07	13.45	12.98	16.12
t-stat	4.52	6.42	4.04	4.58	4.30	4.17	6.44	6.31	6.91	6.31	6.23	7.64

Table A2. CVTURN portfolios after controlling for VOL, TURN, VSHOCK, BA, ILLIQ, RILLIQ.

This table reports double-sorted, value- and equal-weighted portfolios formed daily based on the coefficient of variation of turnover after controlling for volume (VOL), turnover (TURN), volume shocks (VSHOCK), bid-ask spread (BA), Amihud and realized Amihud illiquidity measures (ILLIQ and RILLIQ) between September 1, 2017, and September 1, 2022. For each case, we first sort the cryptocurrencies into three unequal-sized groups (30% lowest, 40% middle, 30% highest) using the control variable, then within each third, we sort cryptocurrencies into quintile portfolios based on the coefficient of variation measures calculated over the 7-day (Panel A) or 30-day (Panel B) rolling windows. The table presents average returns across the three control group to produce quintile portfolios with dispersion in turnover variations but with similar levels of the control variable. It also reports the mean and alpha of the difference in average returns between the lowest and highest variation of turnover portfolios. Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the difference. We report monthly mean and alpha statistics.

	Value-Weighted Portfolios						Equal-Weighted Portfolios					
	VOL	TURN	VSHOCK	BA	ILLIQ	RILLIQ	VOL	TURN	VSHOCK	BA	ILLIQ	RILLIQ
Panel A: CVTURN7												
Low	-2.18	-1.93	-0.78	-3.15	-1.08	-1.30	-2.62	-1.91	-1.96	-2.98	-1.89	-1.62
2	-2.89	-4.23	-0.20	-2.71	-5.93	-6.88	-4.01	-4.59	-3.20	-4.35	-5.17	-4.88
3	-2.56	-4.75	-2.88	-4.87	-3.66	-2.25	-5.70	-6.70	-4.87	-6.18	-5.83	-6.53
4	-8.70	-6.51	-9.63	-8.11	-7.80	-8.24	-9.93	-7.55	-9.42	-8.02	-9.42	-9.34
High	-15.80	-13.95	-12.99	-16.95	-17.44	-17.92	-16.21	-12.37	-12.15	-16.60	-15.65	-15.59
LMH	13.62	12.01	12.21	13.80	16.36	16.62	13.59	10.46	10.19	13.62	13.76	13.97
t-stat	5.54	5.06	4.95	5.53	6.24	6.38	6.93	5.27	5.55	7.04	6.51	6.56
alpha	15.26	13.57	14.42	14.03	16.76	17.25	13.27	10.44	10.56	12.87	12.83	13.06
t-stat	6.43	5.71	5.88	5.61	6.39	6.57	7.14	5.39	5.71	6.73	6.42	6.46
Panel B: CVTURN30												
Low	-5.29	-3.96	0.39	-2.17	-3.27	-3.95	-4.22	-4.86	-2.55	-3.32	-3.51	-3.55
2	-4.25	-3.78	-4.63	-2.51	-3.13	-2.92	-4.26	-4.84	-4.73	-5.48	-4.51	-4.52
3	-5.14	-7.29	-4.83	-7.52	-8.17	-8.27	-5.51	-6.51	-5.77	-6.34	-6.50	-6.36
4	-11.83	-6.94	-7.80	-7.18	-12.15	-12.27	-11.84	-8.22	-8.56	-9.80	-12.58	-12.11
High	-10.06	-9.10	-8.28	-12.37	-9.54	-10.58	-13.05	-8.39	-9.95	-14.10	-11.19	-11.74
LMH	4.76	5.13	8.66	10.19	6.27	6.63	8.83	3.54	7.40	10.78	7.67	8.19
t-stat	1.78	1.91	3.10	3.54	2.15	2.34	4.78	1.84	4.14	5.38	3.87	4.16
alpha	7.52	8.81	11.40	13.15	8.84	9.20	9.43	4.72	7.83	11.08	7.93	8.36
t-stat	2.85	3.54	4.38	4.74	3.23	3.33	4.89	2.48	4.23	5.36	3.88	4.15

Table A3. CVTURN portfolios after controlling for MAX, IVOL, RVOL.

This table reports double-sorted, value- and equal-weighted portfolios formed daily based on the coefficient of variation of turnover after controlling for a maximum daily return over the past 7 and 30 days (MAX7 and MAX30), idiosyncratic volatility (IVOL), realized volatility over the past 7 and 30 days (RVOL7 and RVOL30) between September 1, 2017, and September 1, 2022. For each case, we first sort the cryptocurrencies into three unequal-sized groups (30% lowest, 40% middle, 30% highest) using the control variable, then within each third, we sort cryptocurrencies into quintile portfolios based on the coefficient of variation measures calculated over the 7-day (Panel A) or 30-day (Panel B) rolling windows. The table presents average returns across the three control group to produce quintile portfolios with dispersion in turnover variations but with similar levels of the control variable. It also reports the mean and alpha of the difference in average returns between the lowest and highest variation of turnover portfolios. Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the difference. We report monthly mean and alpha statistics.

	Value-Weighted Portfolios					Equal-Weighted Portfolios				
	MAX7	MAX30	IVOL	RVOL7	RVOL30	MAX7	MAX30	IVOL	RVOL7	RVOL30
Panel A: CVTURN7										
Low	-4.61	-3.69	-5.50	-5.07	-5.34	-2.40	-3.29	-4.25	-2.99	-3.59
2	-3.02	-3.02	-3.89	-4.15	-3.89	-4.64	-4.25	-3.80	-4.40	-4.14
3	-3.51	-3.30	-6.44	-3.63	-6.10	-5.05	-4.85	-7.62	-5.69	-5.80
4	-10.14	-9.15	-11.46	-9.05	-10.31	-10.33	-9.01	-8.92	-9.08	-8.99
High	-16.16	-19.00	-16.18	-15.66	-17.27	-15.71	-17.25	-14.44	-15.74	-16.19
LMH	11.55	15.31	10.68	10.60	11.93	13.31	13.96	10.20	12.76	12.61
t-stat	4.26	5.60	3.97	4.00	4.55	6.68	7.11	5.13	6.60	6.22
alpha	12.94	16.70	11.87	11.76	13.09	12.60	14.15	10.19	11.88	12.44
t-stat	4.99	6.33	4.40	4.46	4.87	6.31	7.09	5.05	6.25	6.10
Panel B: CVTURN30										
Low	-3.04	-4.09	-6.10	-4.22	-5.82	-2.29	-3.84	-4.91	-3.84	-3.89
2	-1.90	-5.56	-7.25	-4.60	-7.13	-5.67	-5.52	-6.12	-4.80	-6.23
3	-7.19	-7.61	-7.21	-7.61	-6.53	-4.52	-7.00	-6.55	-6.40	-7.25
4	-7.42	-5.26	-7.94	-6.67	-6.46	-11.32	-9.31	-8.48	-9.12	-7.51
High	-12.95	-12.56	-13.57	-13.17	-13.86	-15.08	-13.39	-13.48	-14.77	-14.21
LMH	9.91	8.47	7.46	8.94	8.04	12.79	9.55	8.56	10.93	10.32
t-stat	3.40	2.98	2.76	3.03	2.94	6.49	4.75	4.32	5.46	5.15
alpha	12.74	12.51	9.83	11.97	10.81	13.41	10.53	8.97	11.67	10.92
t-stat	4.46	4.61	3.63	4.32	4.02	6.69	5.26	4.46	5.86	5.53

Table A4. CVTURN portfolios after controlling for NEW, ACT, VAR95, ES95.

This table reports double-sorted, value- and equal-weighted portfolios formed daily based on the coefficient of variation of turnover after controlling for a sum of new or active addresses over the past 7 and 30 days (NEW7, NEW30, ACT7, and ACT30), value-at-risk (VAR95), and expected shortfall (ES95) between September 1, 2017, and September 1, 2022. For each case, we first sort the cryptocurrencies into three unequal-sized groups (30% lowest, 40% middle, 30% highest) using the control variable, then within each third, we sort cryptocurrencies into quintile portfolios based on the coefficient of variation measures calculated over the 7-day (Panel A) or 30-day (Panel B) rolling windows. The table presents average returns across the three control group to produce quintile portfolios with dispersion in turnover variations but with similar levels of the control variable. It also reports the mean and alpha of the difference in average returns between the lowest and highest variation of turnover portfolios. Alphas are calculated after adjusting for the market, size, and momentum factors of [Liu, Tsyvinski, and Wu \(2022\)](#). [Newey and West \(1986\)](#) adjusted t-statistics are presented for alpha estimates and the mean of the difference. We report monthly mean and alpha statistics.

	Value-Weighted Portfolios						Equal-Weighted Portfolios					
	NEW7	NEW30	ACT7	ACT30	VAR95	ES95	NEW7	NEW30	ACT7	ACT30	VAR95	ES95
Panel A: CVTURN7												
Low	-2.65	-3.11	-2.60	-3.21	-3.20	-3.16	-1.68	-2.42	-1.87	-2.23	-3.10	-2.62
2	-1.36	-2.19	-0.71	-1.88	-6.69	-7.17	-2.85	-2.75	-2.75	-2.51	-3.95	-4.54
3	-3.04	-2.88	-3.26	-4.42	-5.31	-3.65	-4.98	-5.14	-5.24	-6.38	-4.57	-4.05
4	-7.91	-7.61	-8.65	-7.09	-11.17	-10.34	-8.80	-8.32	-8.11	-7.22	-8.11	-8.40
High	-16.50	-18.10	-17.29	-18.50	-15.66	-14.47	-18.11	-17.90	-17.88	-17.69	-14.80	-15.46
LMH	13.85	14.99	14.69	15.29	12.46	11.31	16.43	15.48	16.01	15.45	11.71	12.84
t-stat	5.87	6.56	6.05	6.55	4.64	4.32	8.85	8.78	8.52	8.47	6.22	6.99
alpha	14.99	16.69	15.05	16.67	13.81	12.24	16.76	16.10	15.85	15.65	12.05	12.94
t-stat	6.46	7.15	6.34	6.88	5.01	4.59	8.96	9.08	8.43	8.45	6.47	7.21
Panel B: CVTURN30												
Low	-2.11	-2.10	-3.27	-2.11	-4.35	-4.98	-2.85	-3.19	-3.21	-2.87	-3.32	-3.51
2	-4.11	-3.47	-3.92	-4.57	-7.74	-6.68	-3.98	-3.77	-3.83	-4.56	-5.03	-5.44
3	-5.46	-4.21	-4.70	-3.79	-6.14	-7.18	-5.76	-5.22	-5.94	-5.44	-5.69	-5.33
4	-8.46	-8.48	-8.10	-8.89	-7.49	-6.39	-8.38	-8.00	-6.94	-8.53	-7.44	-8.02
High	-14.42	-16.12	-14.93	-14.44	-13.25	-13.58	-16.07	-16.61	-16.24	-14.90	-13.45	-13.07
LMH	12.31	14.02	11.66	12.34	8.90	8.59	13.22	13.43	13.03	12.03	10.13	9.56
t-stat	4.44	4.86	4.37	4.44	3.27	3.24	6.71	6.43	6.60	5.93	5.02	4.90
alpha	14.20	16.27	13.67	14.39	11.03	11.02	14.15	14.29	13.90	12.88	10.90	10.09
t-stat	5.39	6.01	5.27	5.65	4.30	4.33	7.13	6.80	7.06	6.31	5.41	5.07

Table A5. Cryptocurrency-level cross-sectional regressions.

This table reports daily [Fama and MacBeth \(1973\)](#) results for cross-sectional regressions between September 1, 2017, and September 1, 2022. Each day, we estimate a cross-sectional regression of one-day-ahead cryptocurrency returns on the coefficient of variation of turnover and control variables, one at a time. We then report the time-series averages of the estimates, their [Newey and West \(1986\)](#) adjusted t-statistics (cells in the grey font), and average R-squared statistics. The last row in each panel reports the monthly CVVOL risk premium, which is calculated as the product of the CVVOL coefficient and the spread in median CVVOL characteristics between the lowest and highest CVVOL quintiles. The table shows results based on the coefficient of variation measures calculated over the 7-day (Panel A) or 30-day (Panel B) rolling windows. The control variables include CAPM beta (BETA); market capitalization (SIZE); past 7- and 30-day momentum (MOM7 and MOM30), network-to-market ratio (BM); turnover (TURN); volume shocks (VSHOCK); bid-ask spread (Bid-Ask); Amihud and realized illiquidity measures (ILLIQ and RILLIQ); a maximum daily return over the past 7 and 30 days (MAX7 and MAX30), idiosyncratic volatility (IVOL), realized volatility over the past 7 and 30 days (RVOL7 and RVOL30), value-at-risk (VAR95), and expected shortfall (ES95).

Panel A: 7-day look-back period															
	CVTURN7	BETA	SIZE	MOM7	BM	TURN	VSHOCK	BA	ILLIQ	RILLIQ	MAX7	IVOL	RVOL7	Var95	ES95
Intercept	0.0003	0.0008	0.0002	-0.0007	0.0002	0.0002	-0.0003	0.0032	0.0003	0.0003	0.0018	0.0029	0.0030	0.0043	0.0038
CVTURN7	0.23	0.62	0.17	-0.50	0.13	0.16	-0.20	2.27	0.22	0.20	1.33	2.27	2.27	3.19	2.75
	-0.0043	-0.0043	-0.0044	-0.0037	-0.0044	-0.0043	-0.0030	-0.0025	-0.0040	-0.0040	-0.0025	-0.0026	-0.0023	-0.0029	-0.0032
CONTROL	-7.85	-8.54	-7.91	-7.17	-7.96	-7.63	-5.88	-4.24	-7.49	-7.47	-4.37	-4.71	-4.08	-5.21	-5.79
	-0.0005	0.0170	0.0170	-0.0378	0.0162	0.0050	-0.0003	-0.0664	0.0002	0.0058	-0.0249	-0.0552	-0.0521	-0.0390	-0.0238
R^2_{adj}	0.92%	-0.68	2.50	-2.33	0.81	2.88	-1.25	-8.31	0.05	0.29	-5.23	-5.31	-5.90	-5.92	-5.20
		3.74%	0.10%	4.51%	0.54%	0.19%	1.41%	5.85%	5.06%	5.07%	5.19%	4.92%	5.57%	4.91%	4.50%
RP(CVTURN7)	-12.65	-12.73	-12.82	-10.90	-12.76	-12.49	-8.77	-7.42	-11.72	-11.63	-7.22	-7.61	-6.86	-8.46	-9.41
Panel B: 30-day look-back period															
	CVTURN30	BETA	SIZE	MOM30	BM	TURN	VSHOCK	BA	ILLIQ	RILLIQ	MAX30	IVOL	RVOL30	Var95	ES95
Intercept	-0.0004	0.0000	-0.0004	-0.0005	-0.0004	-0.0006	-0.0010	0.0028	-0.0006	-0.0006	0.0015	0.0021	0.0033	0.0038	0.0032
CVTURN30	-0.28	0.00	-0.28	-0.42	-0.35	-0.45	-0.79	2.05	-0.47	-0.47	1.20	1.65	2.51	2.77	2.37
	-0.0023	-0.0022	-0.0024	-0.0021	-0.0024	-0.0021	-0.0012	-0.0013	-0.0018	-0.0018	-0.0011	-0.0011	-0.0010	-0.0012	-0.0014
CONTROL	-5.48	-5.46	-5.54	-4.83	-5.41	-5.20	-3.09	-2.93	-4.29	-4.27	-2.41	-2.19	-2.10	-2.73	-3.26
	-0.0004	0.0179	0.0179	-0.0634	0.0289	0.0056	-0.0003	-0.0682	0.0003	0.0048	-0.0150	-0.0602	-0.0630	-0.0401	-0.0247
R^2_{adj}	0.75%	-0.56	2.93	-1.69	1.74	3.47	-1.29	-8.46	0.06	0.20	-4.73	-5.65	-5.47	-6.06	-5.39
		3.61%	-0.09%	3.51%	0.36%	0.00%	1.22%	5.72%	4.94%	4.95%	3.80%	4.75%	4.88%	4.73%	4.29%
RP(CVTURN30)	-8.76	-8.67	-9.24	-8.06	-9.11	-8.13	-4.55	-4.92	-6.87	-6.84	-4.37	-4.14	-3.94	-4.48	-5.34